

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

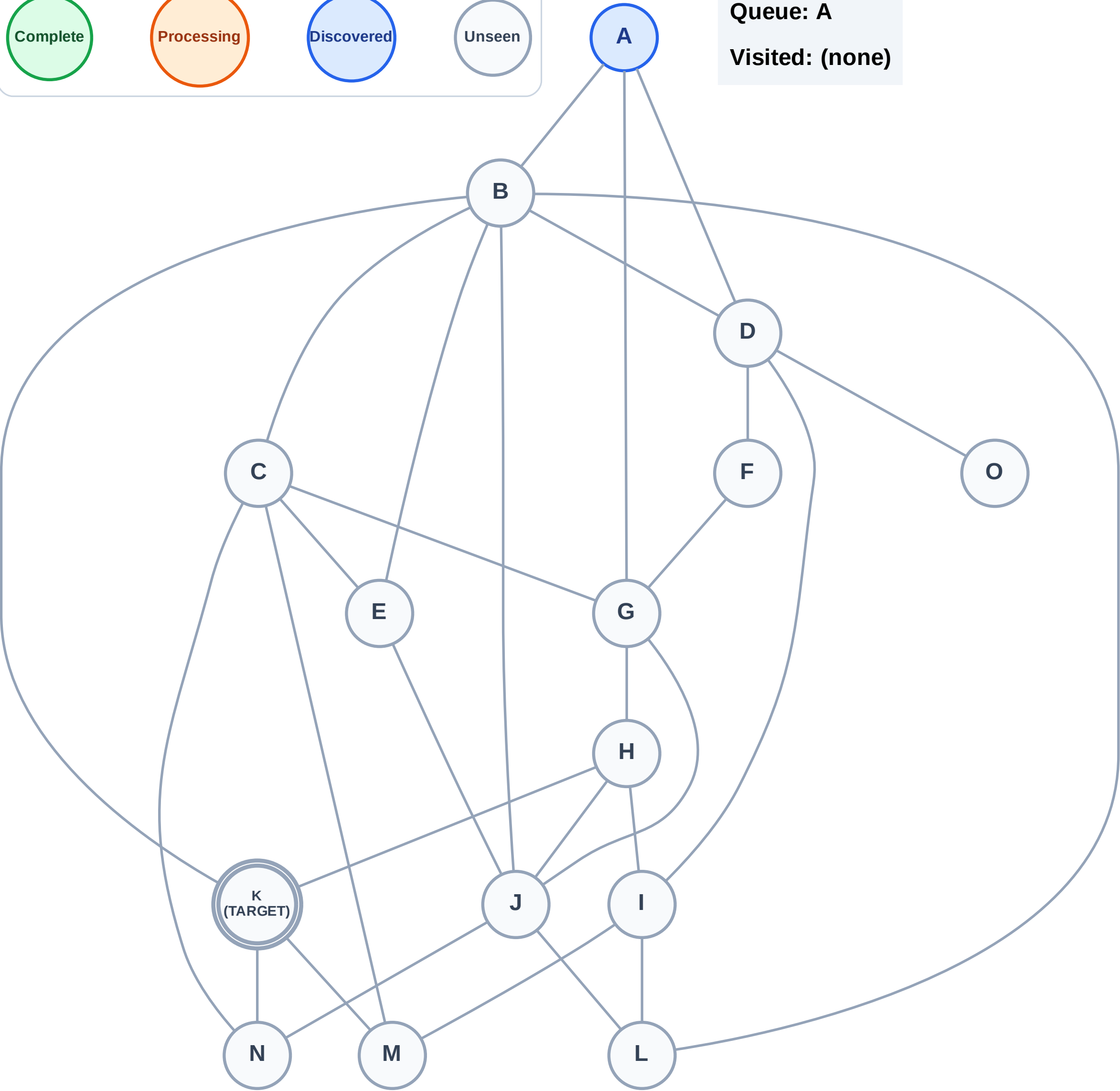
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

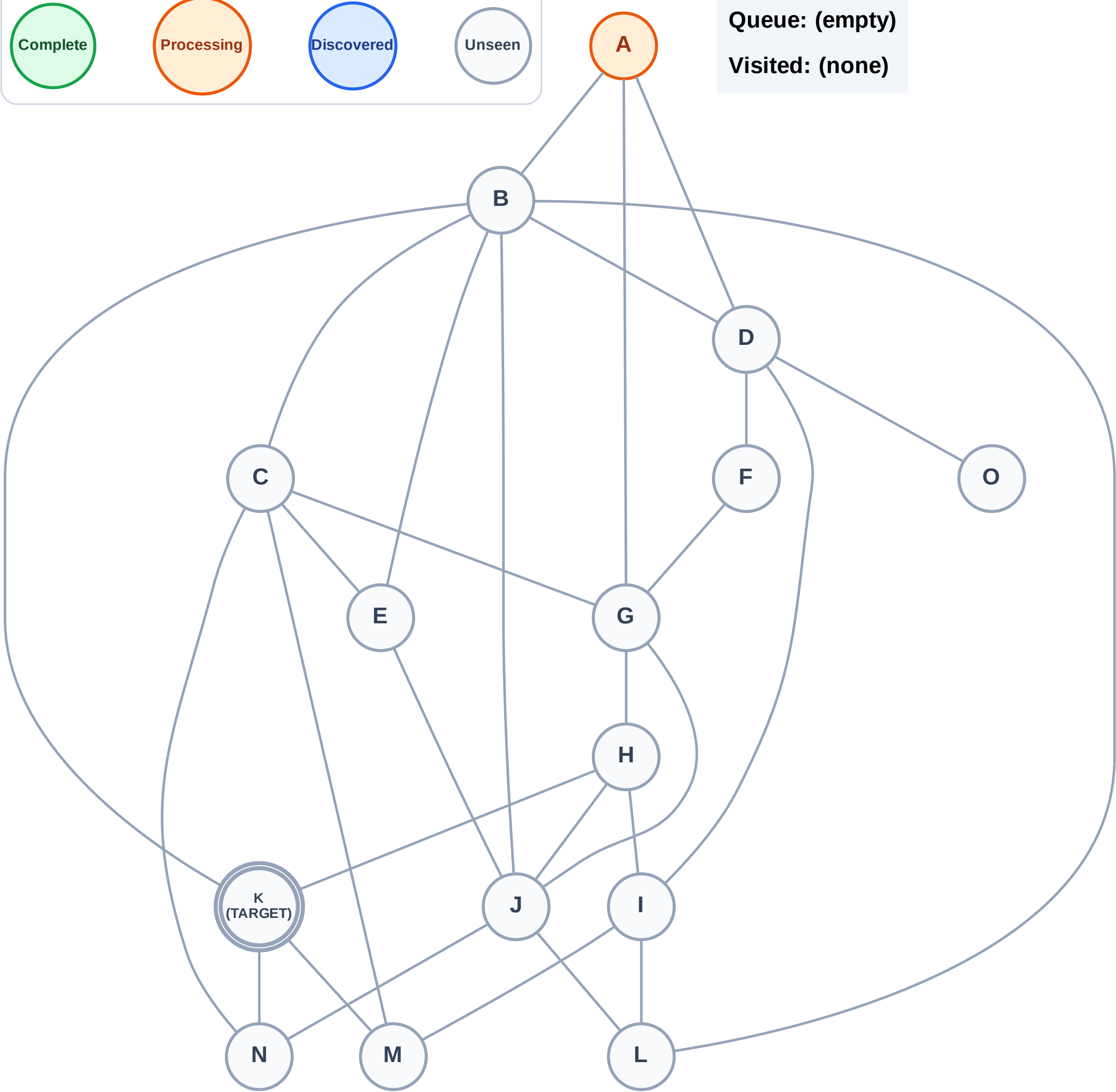
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, D, G from A

Legend



Complete



Processing



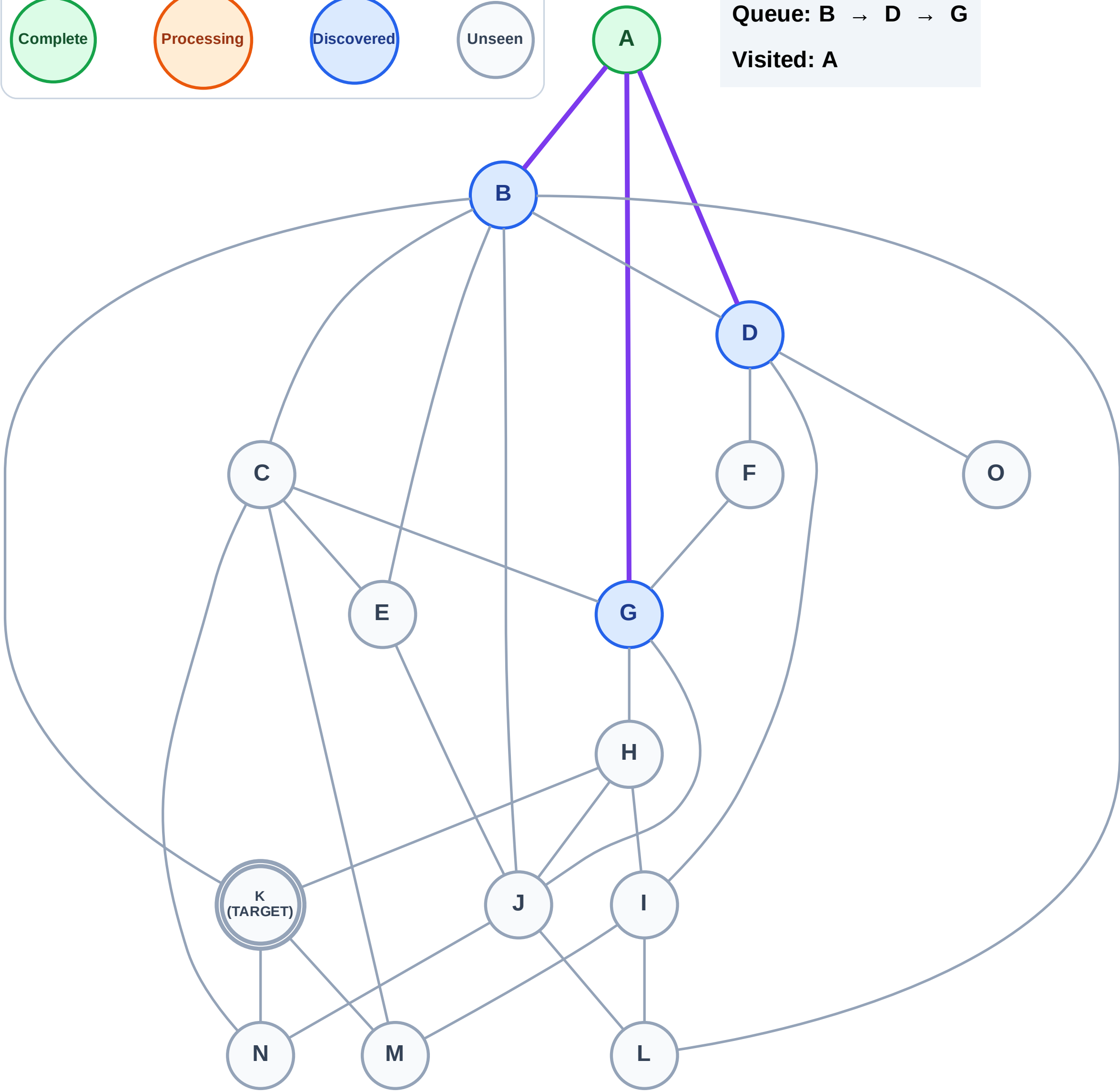
Discovered



Unseen

Queue: B → D → G

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

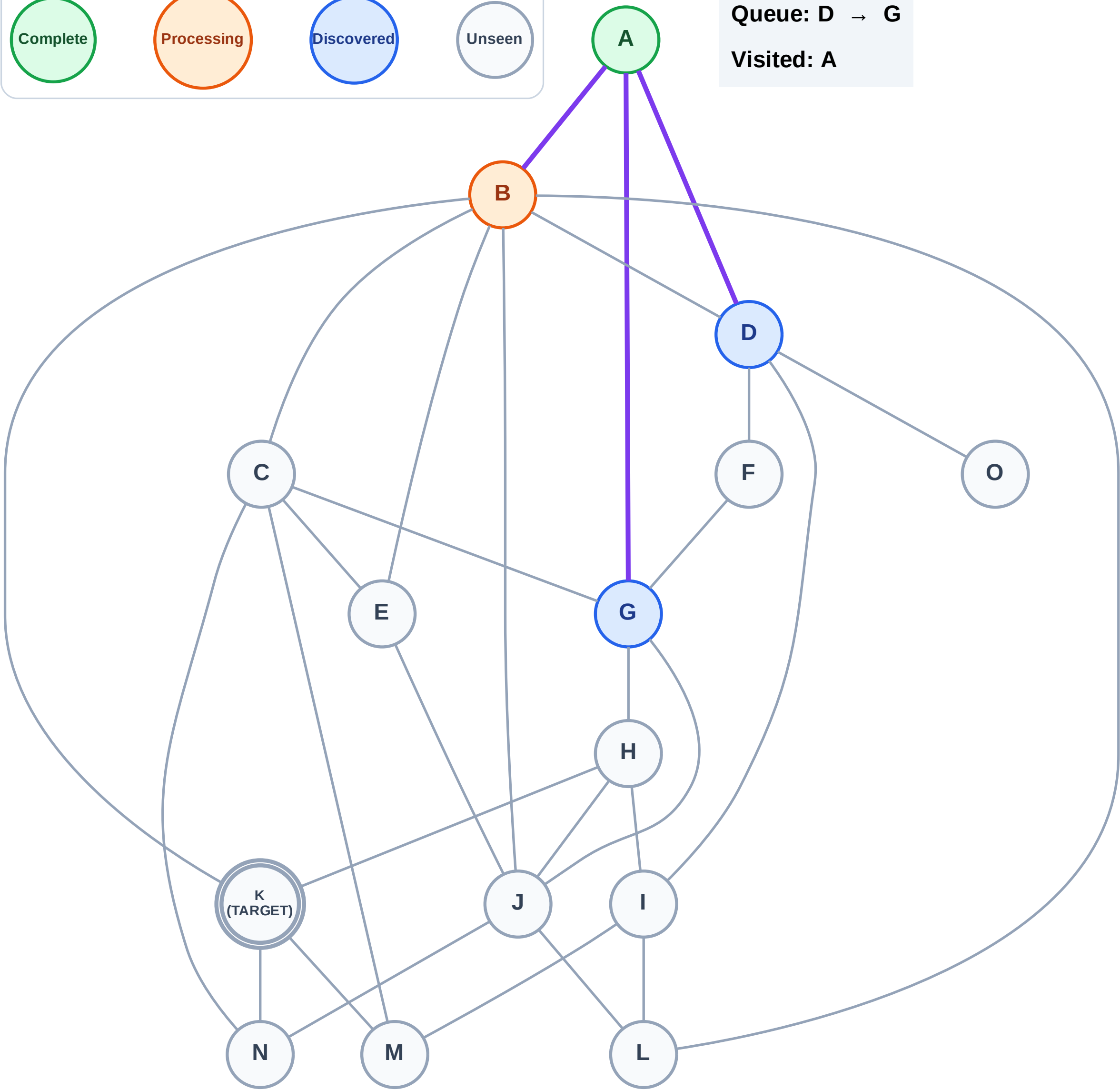
Processing

Discovered

Unseen

Queue: D → G

Visited: A



BFS Traversal

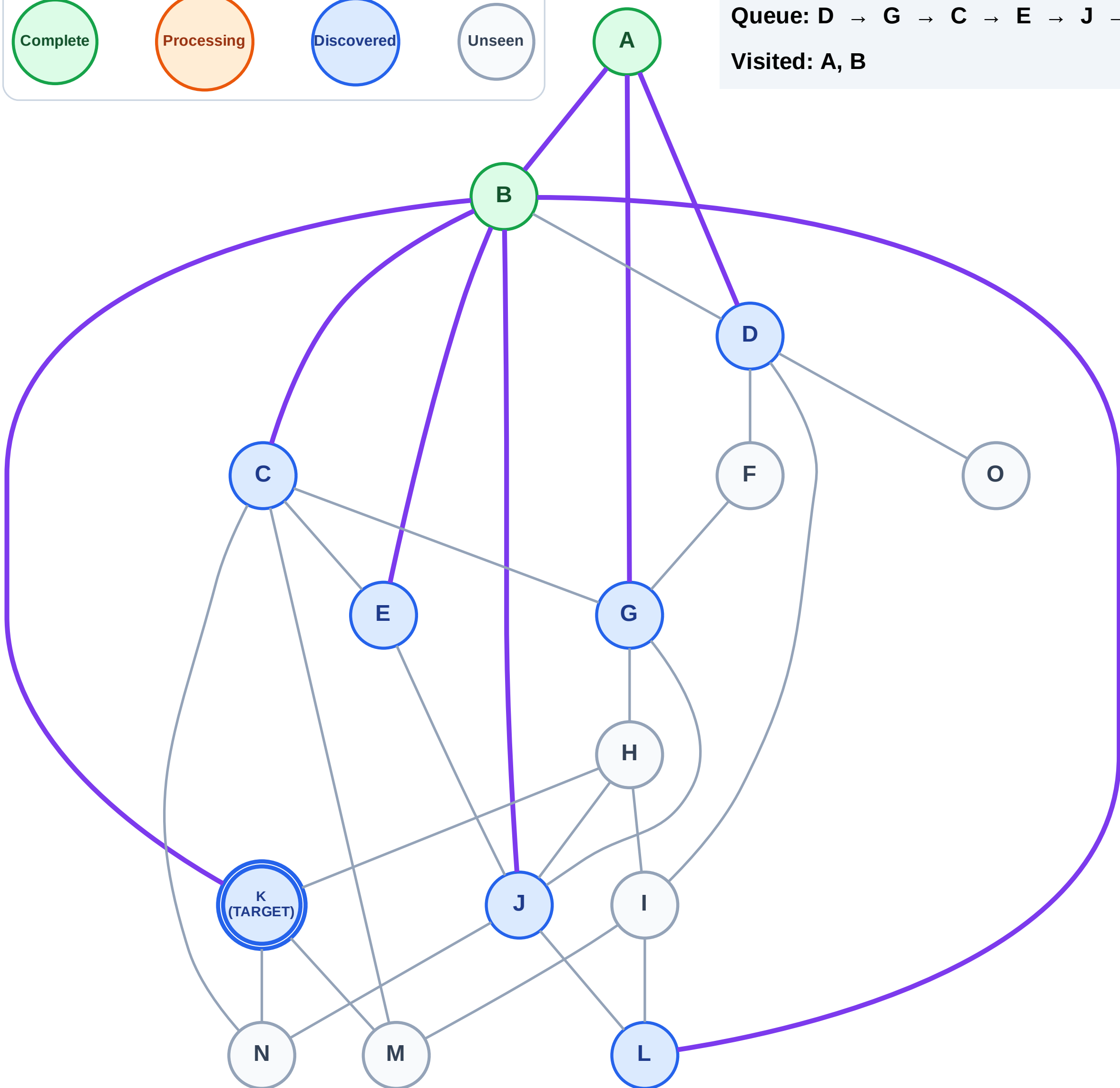
Step 5: Discover C, E, J, K, L from B

Legend



Queue: D → G → C → E → J → K → L

Visited: A, B



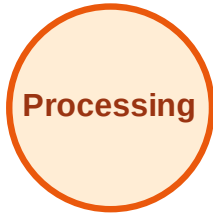
BFS Traversal

Step 6: Process node D

Legend



Complete



Processing



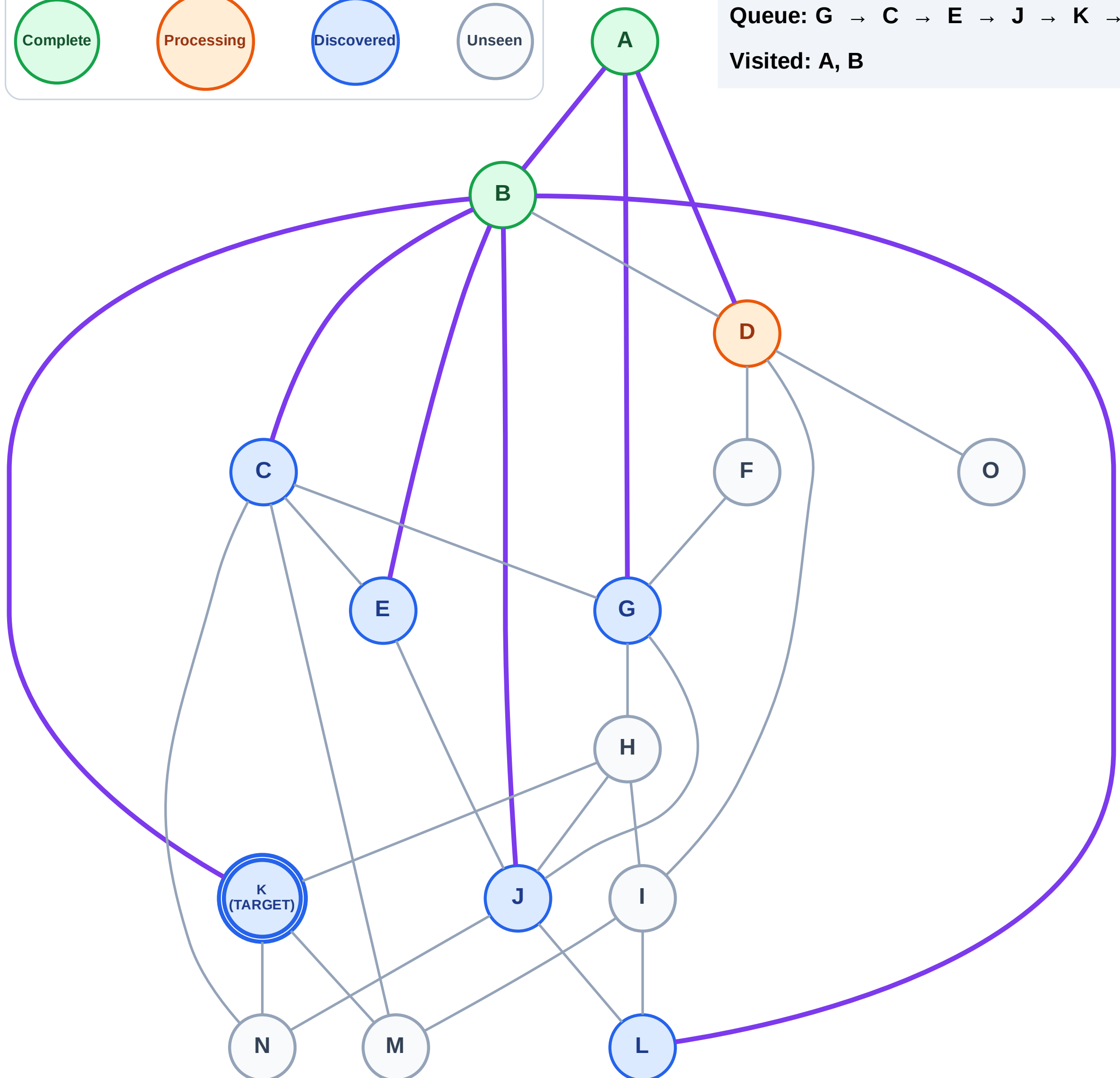
Discovered



Unseen

Queue: G → C → E → J → K → L

Visited: A, B



BFS Traversal

Step 7: Discover F, I, O from D

Legend

Complete

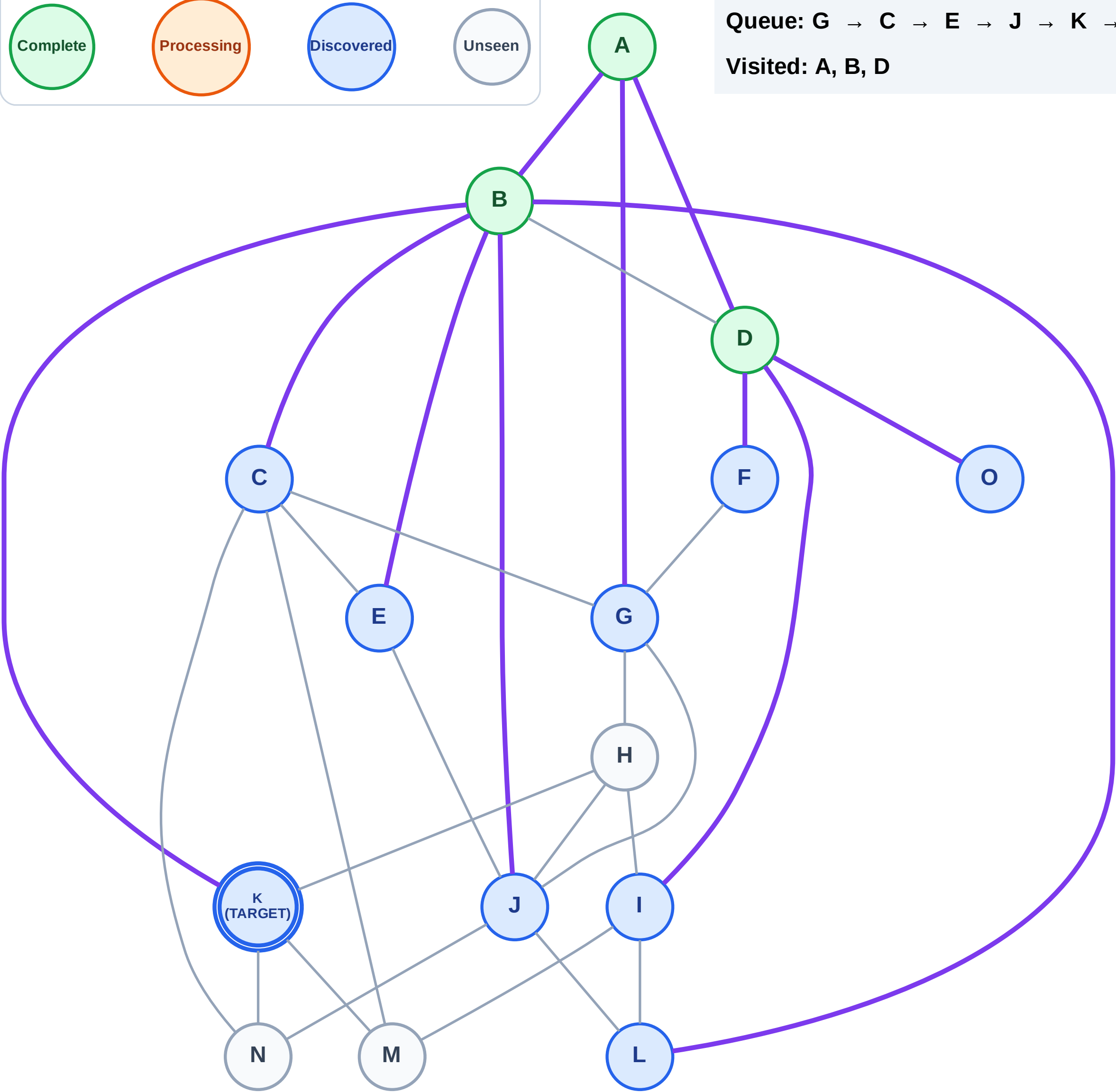
Processing

Discovered

Unseen

Queue: G → C → E → J → K → L → F → I → O

Visited: A, B, D



BFS Traversal

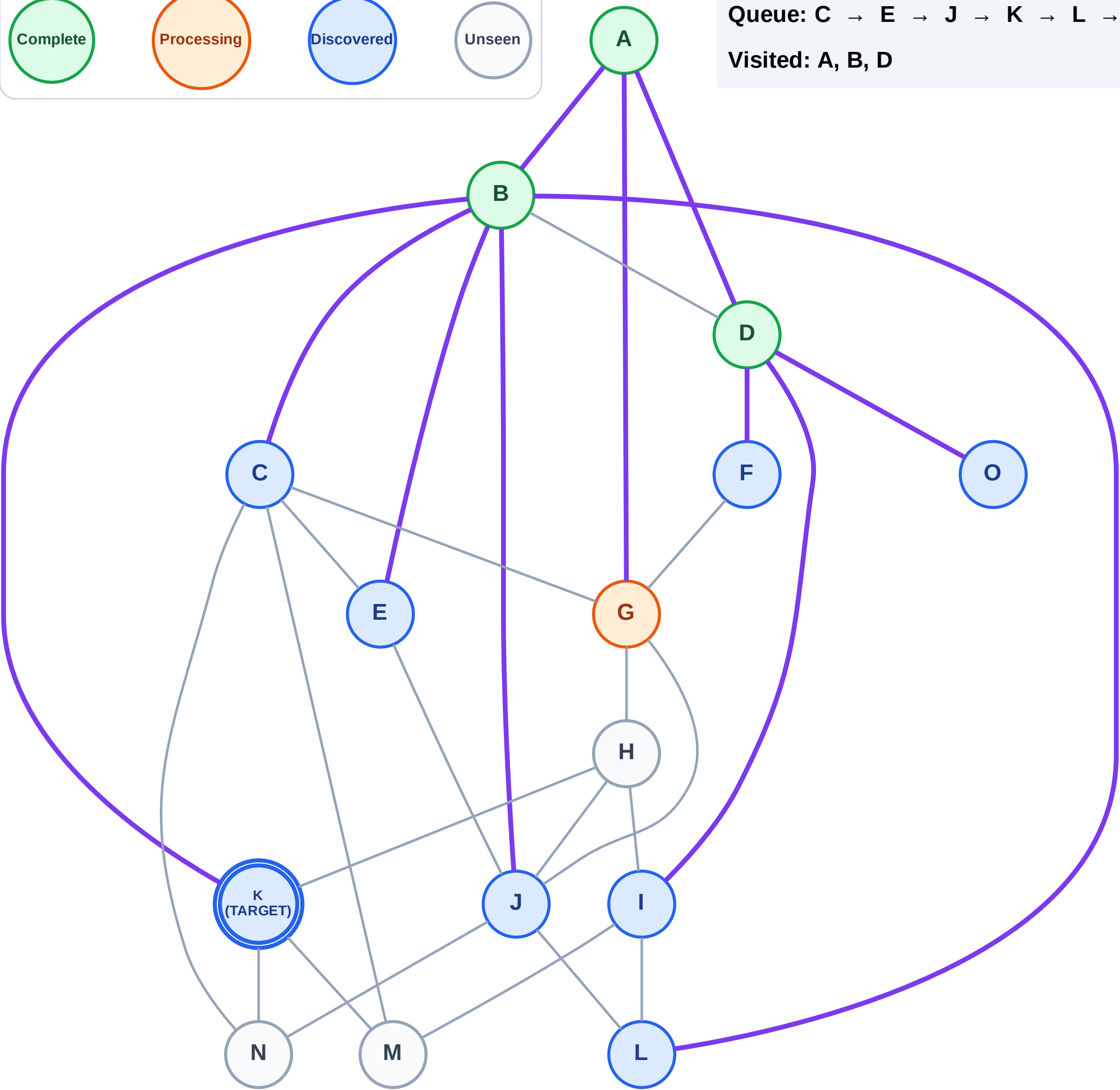
Step 8: Process node G

Legend



Queue: C → E → J → K → L → F → I → O

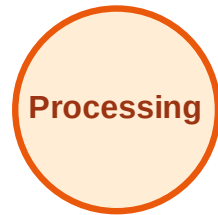
Visited: A, B, D



BFS Traversal

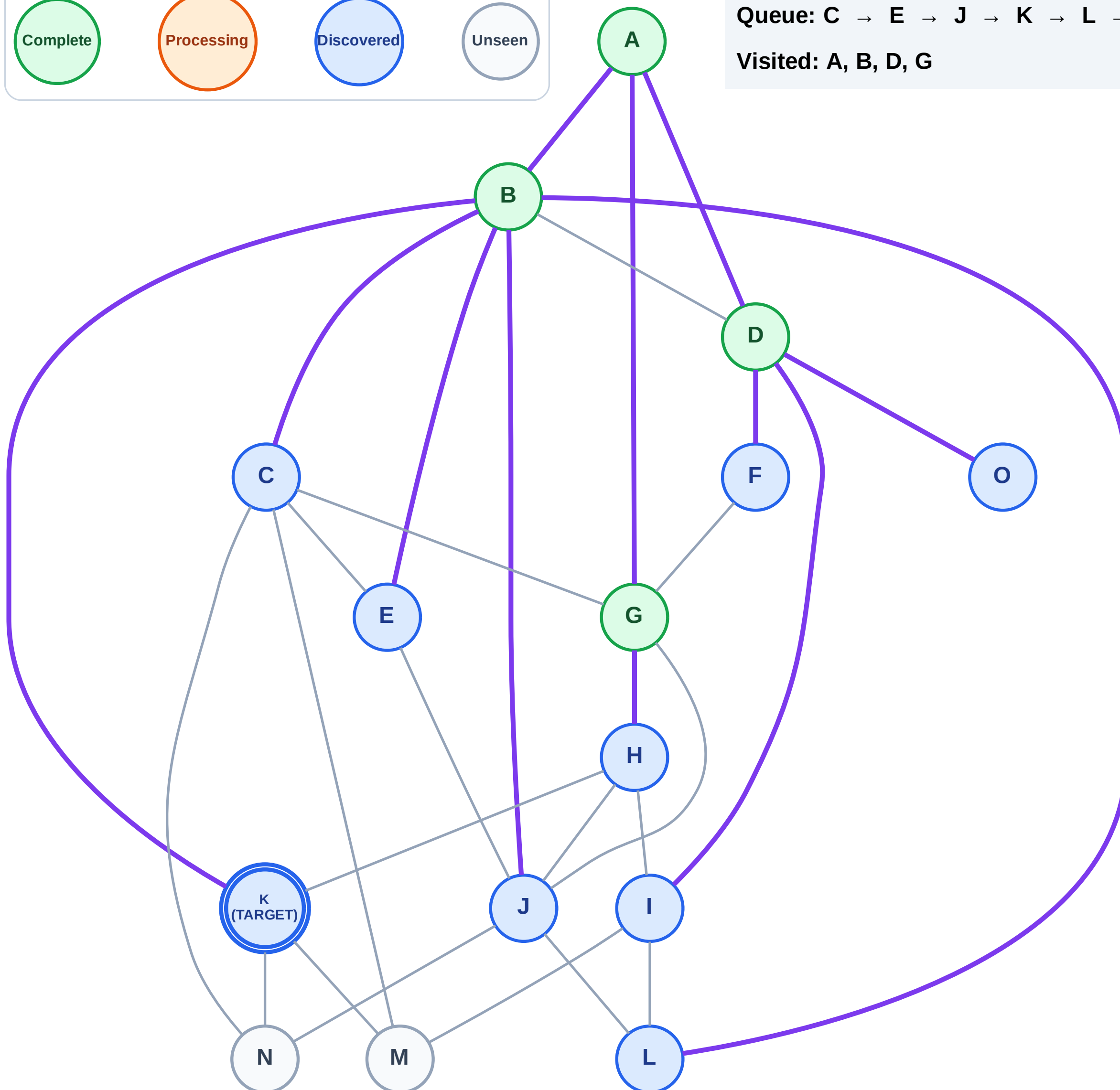
Step 9: Discover H from G

Legend



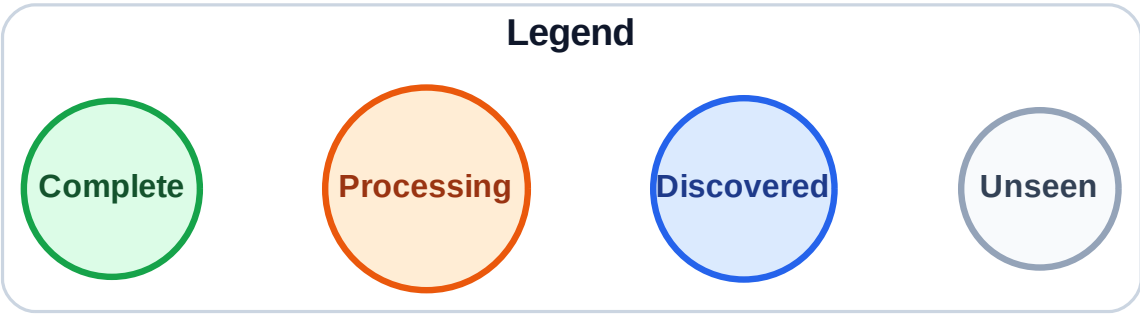
Queue: C → E → J → K → L → F → I → O → H

Visited: A, B, D, G



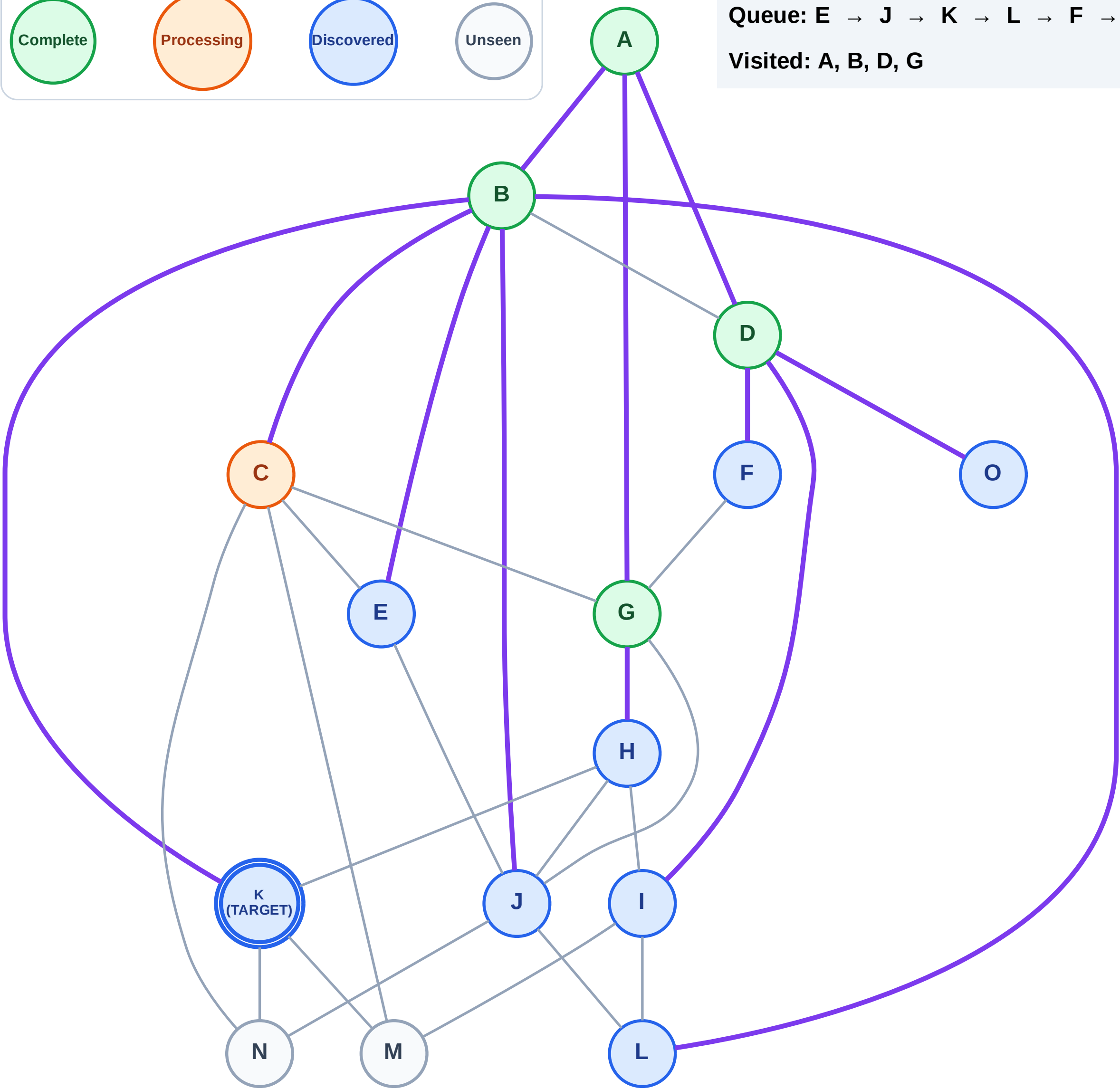
BFS Traversal

Step 10: Process node C



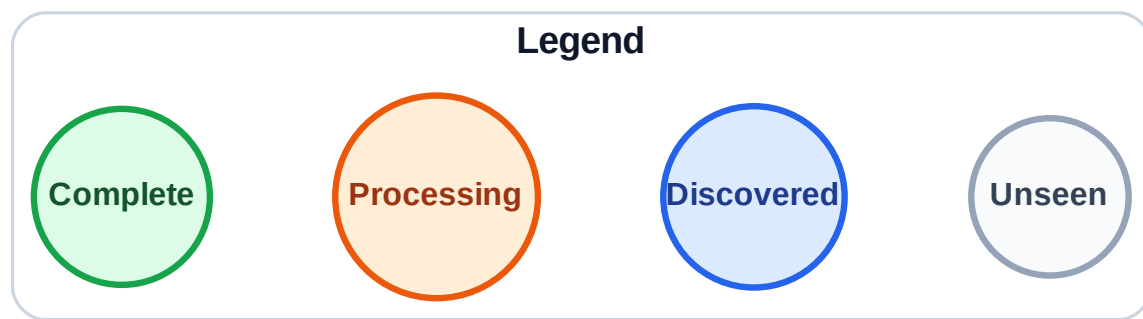
Queue: E → J → K → L → F → I → O → H

Visited: A, B, D, G



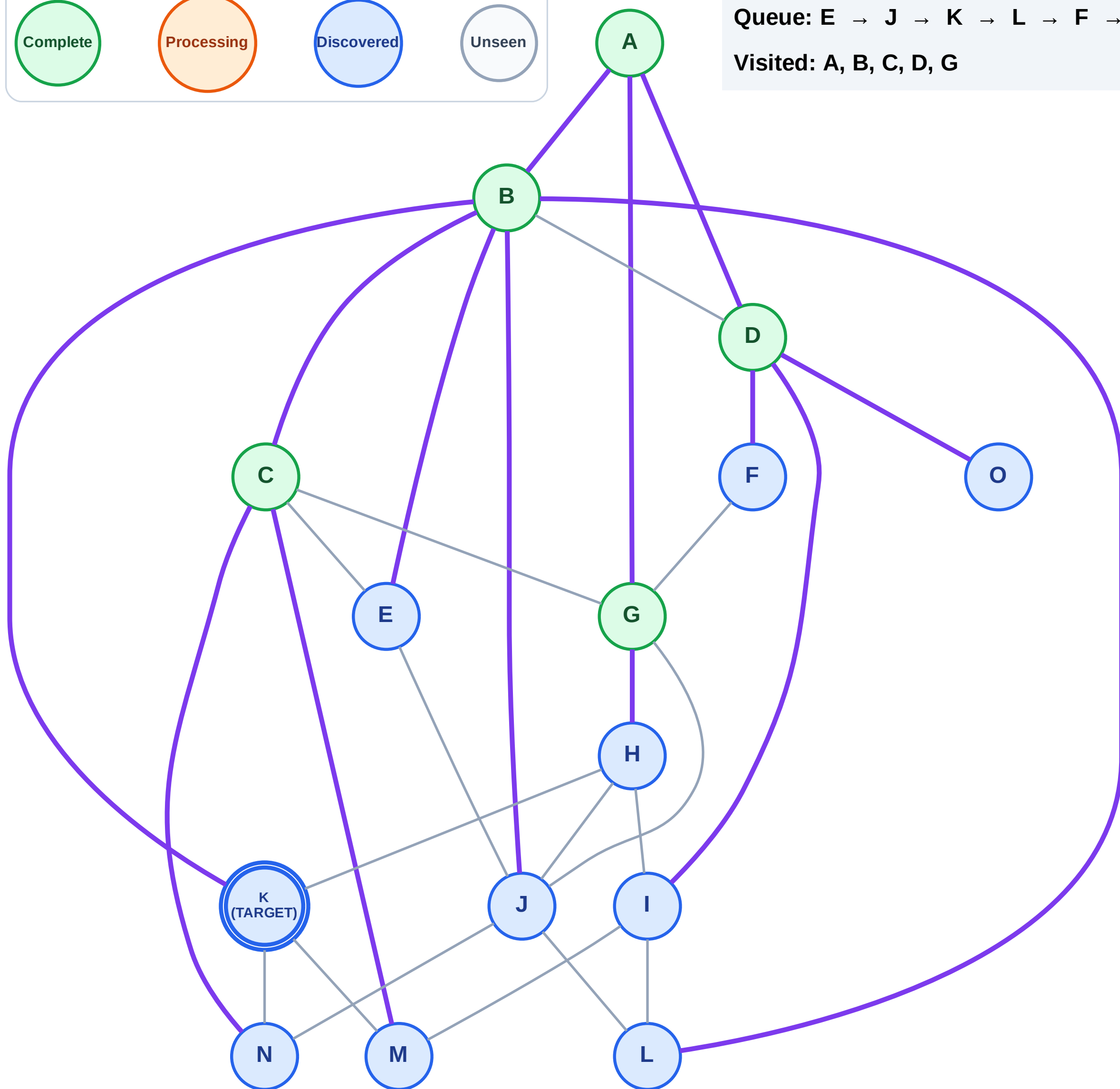
BFS Traversal

Step 11: Discover M, N from C



Queue: E → J → K → L → F → I → O → H → M → N

Visited: A, B, C, D, G



BFS Traversal

Step 12: Process node E

Legend

Complete

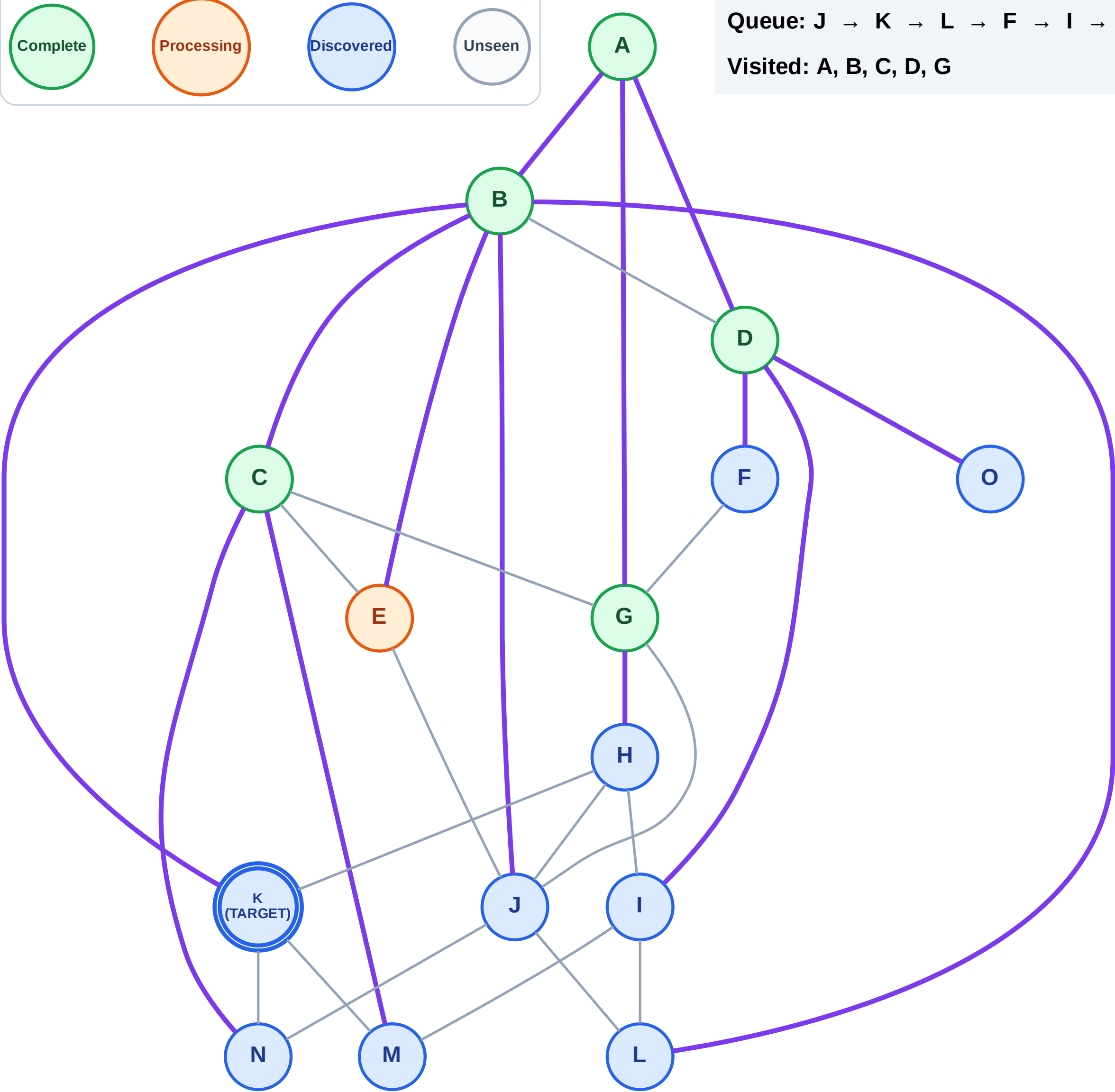
Processing

Discovered

Unseen

Queue: J → K → L → F → I → O → H → M → N

Visited: A, B, C, D, G



BFS Traversal

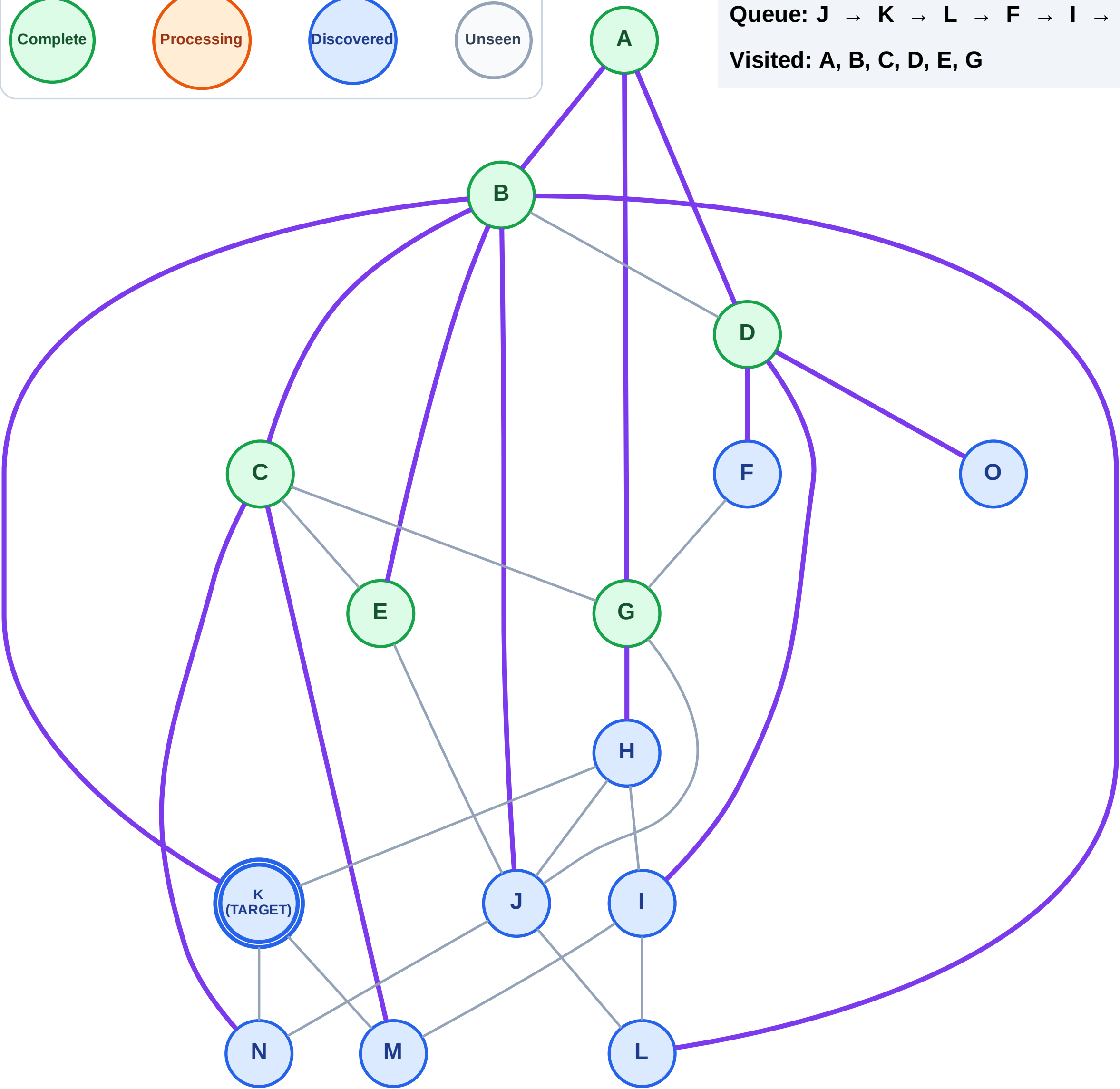
Step 13: No new neighbors from E

Legend



Queue: J → K → L → F → I → O → H → M → N

Visited: A, B, C, D, E, G



BFS Traversal

Step 14: Process node J

Legend

Complete

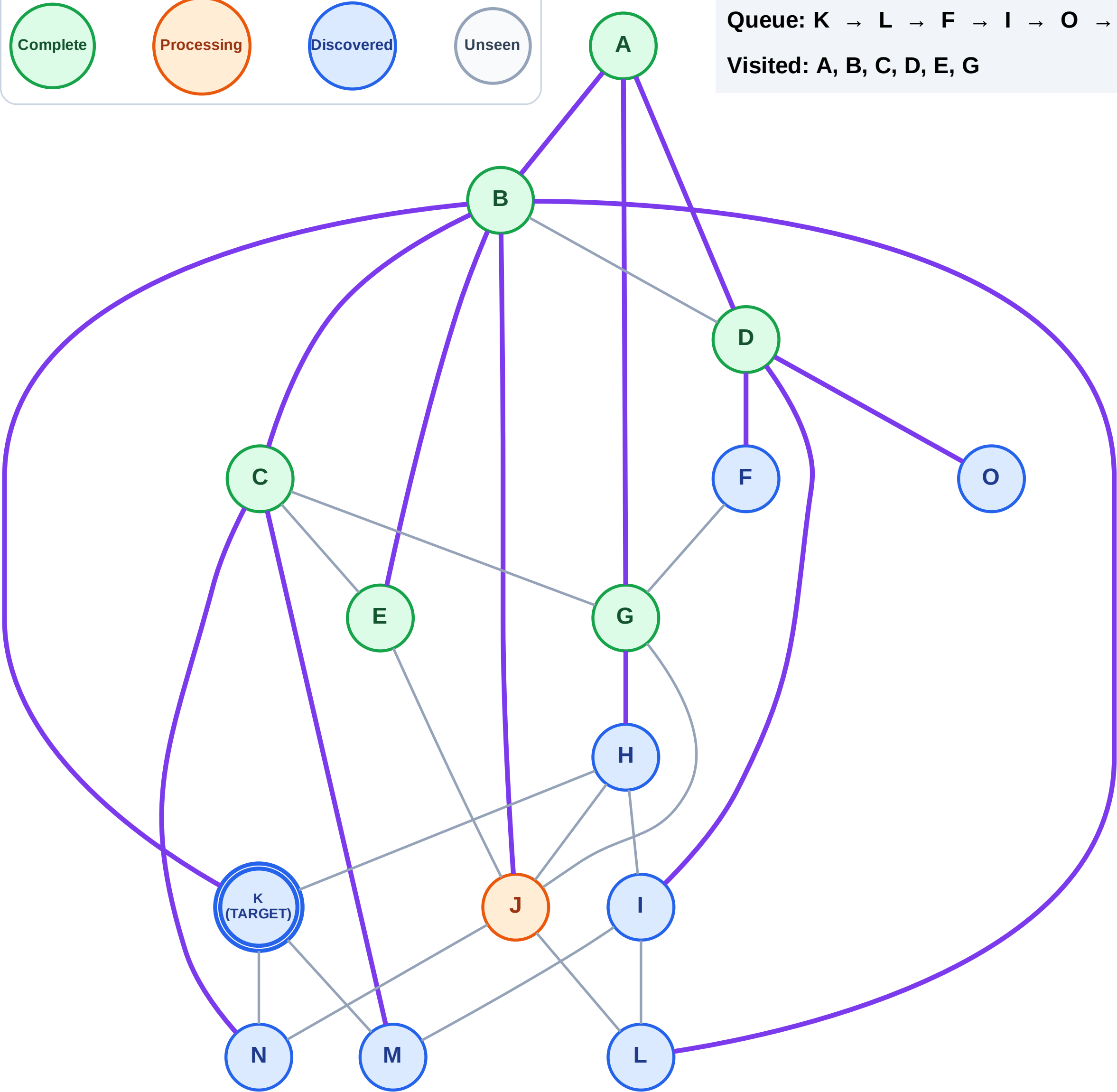
Processing

Discovered

Unseen

Queue: K → L → F → I → O → H → M → N

Visited: A, B, C, D, E, G



BFS Traversal

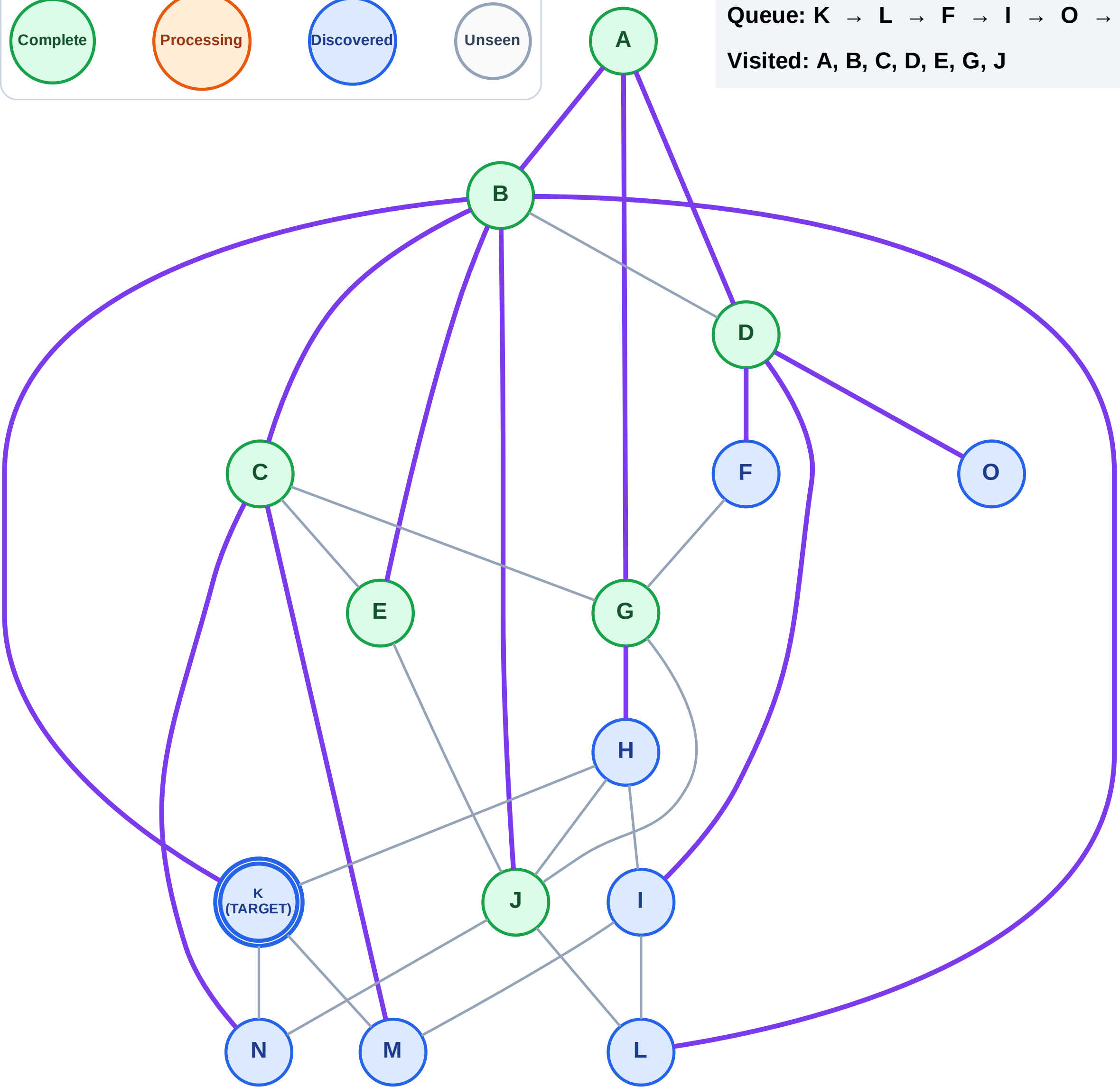
Step 15: No new neighbors from J

Legend



Queue: K → L → F → I → O → H → M → N

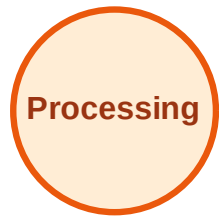
Visited: A, B, C, D, E, G, J



BFS Traversal

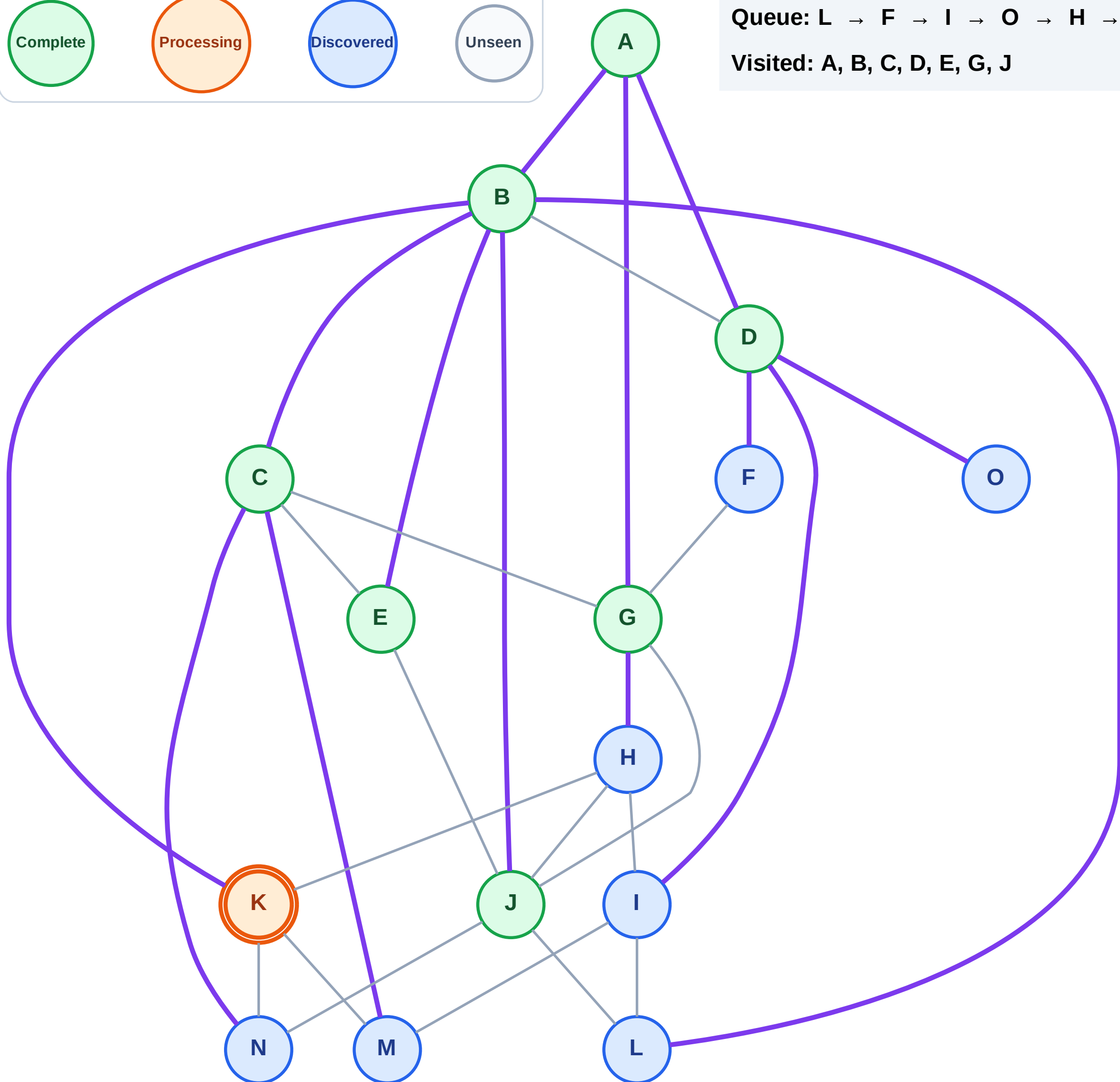
Step 16: Process node K

Legend



Queue: L → F → I → O → H → M → N

Visited: A, B, C, D, E, G, J



BFS Traversal

Step 17: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

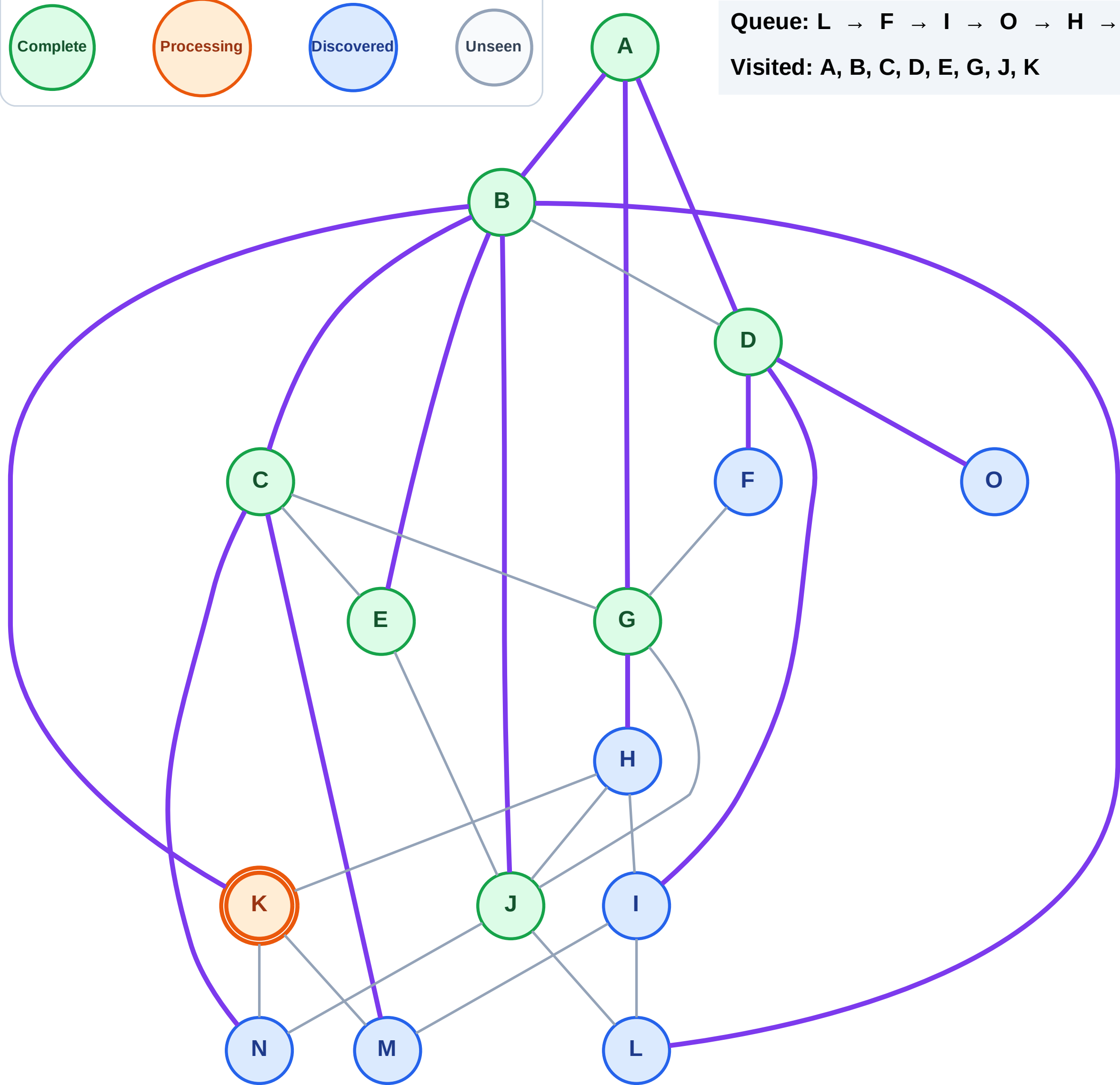
Processing

Discovered

Unseen

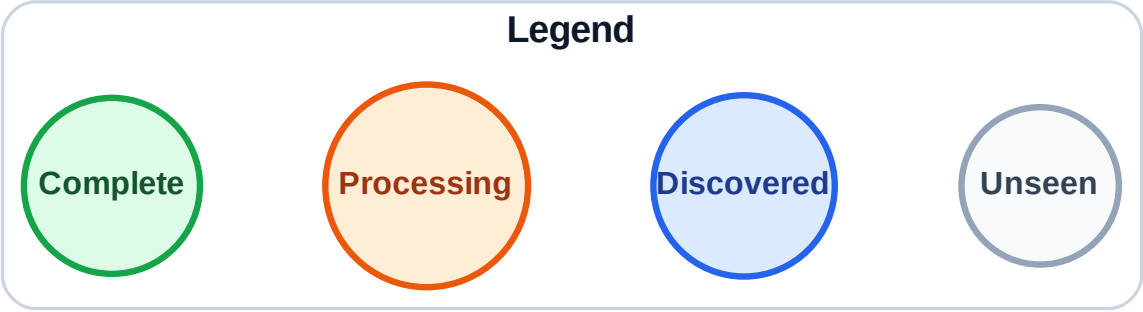
Queue: L → F → I → O → H → M → N

Visited: A, B, C, D, E, G, J, K

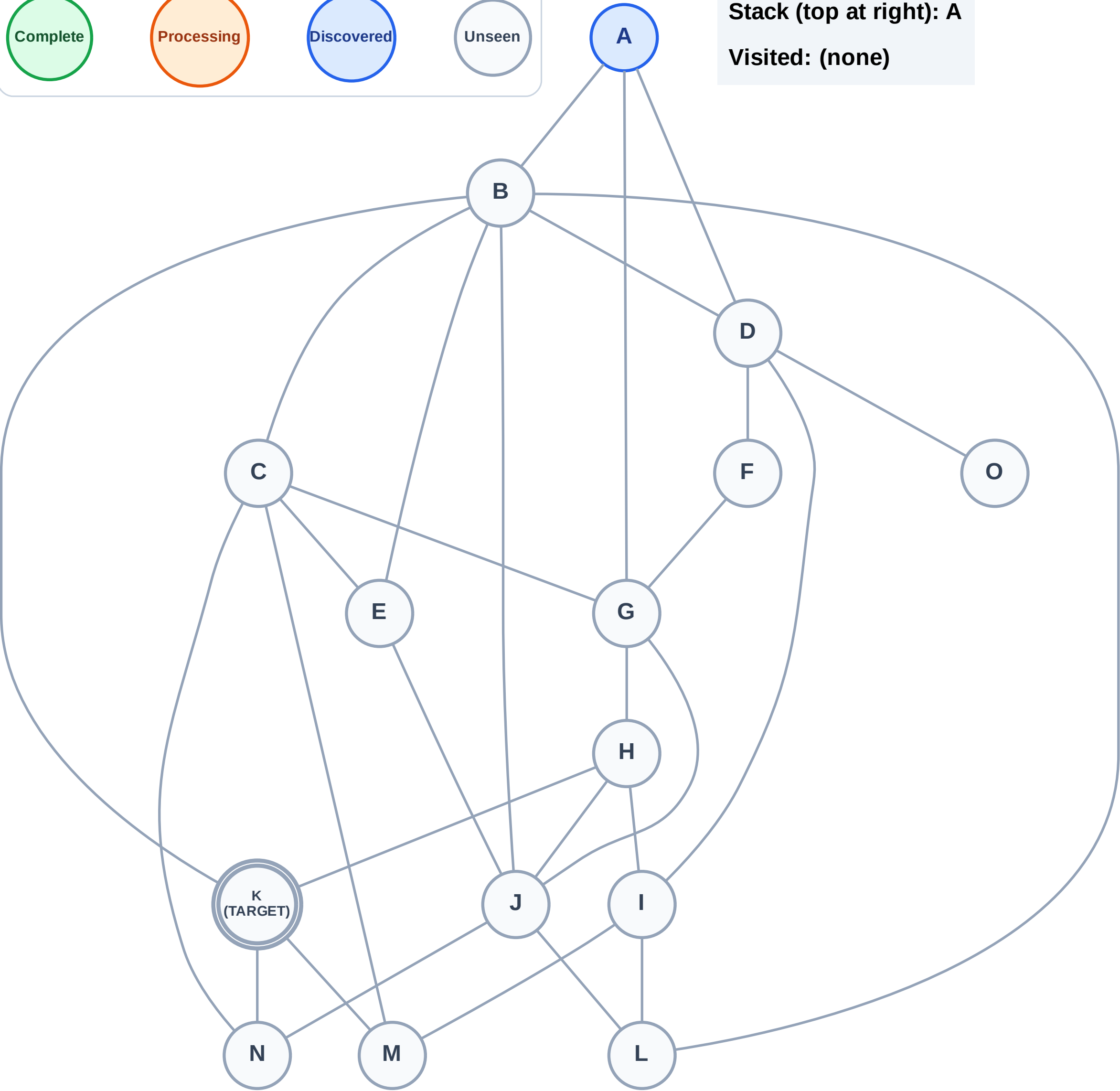


DFS Traversal

Step 1: Start at A (Seeking target node K)



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

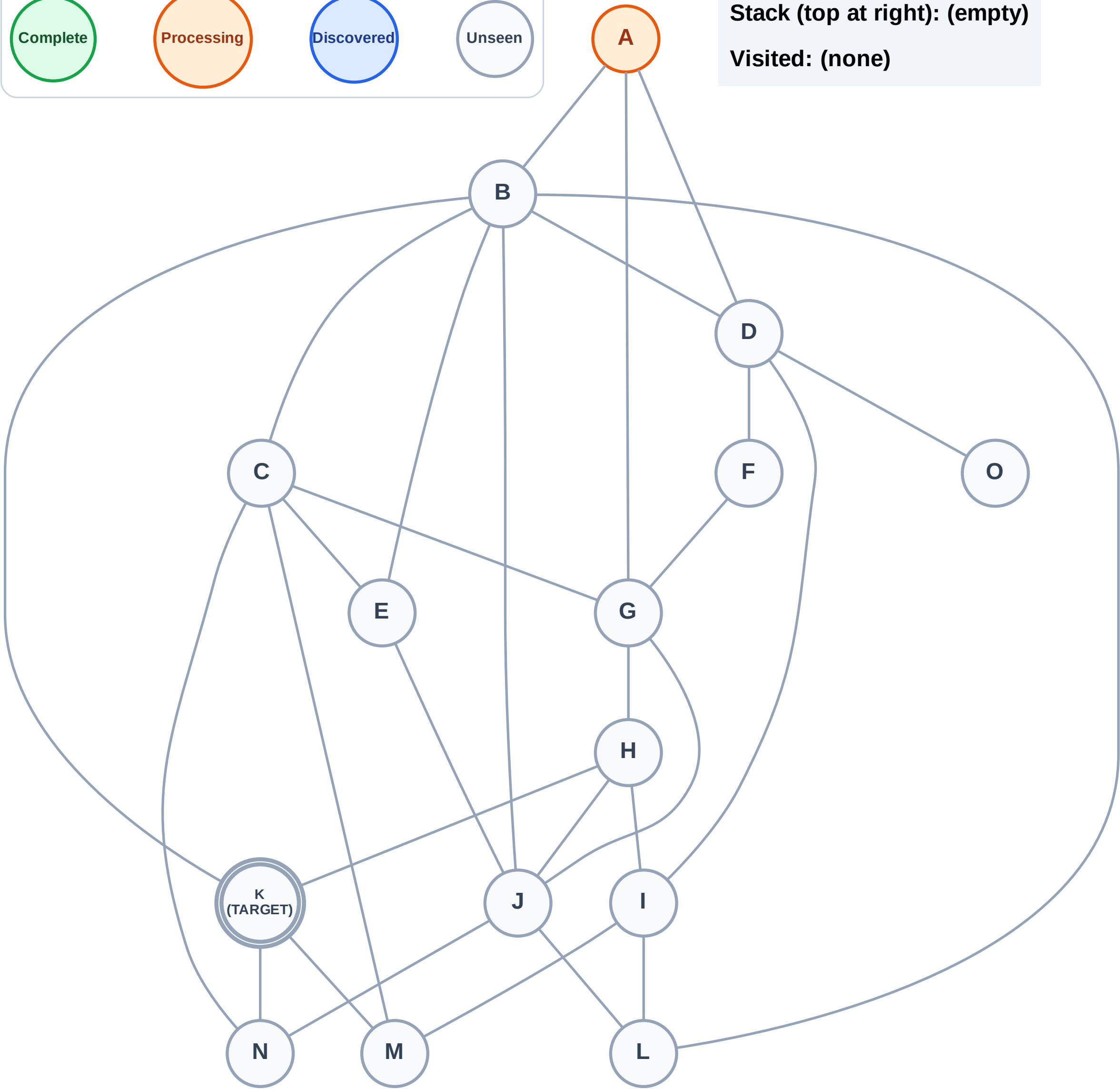
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover G, D, B from A

Legend



Complete



Processing



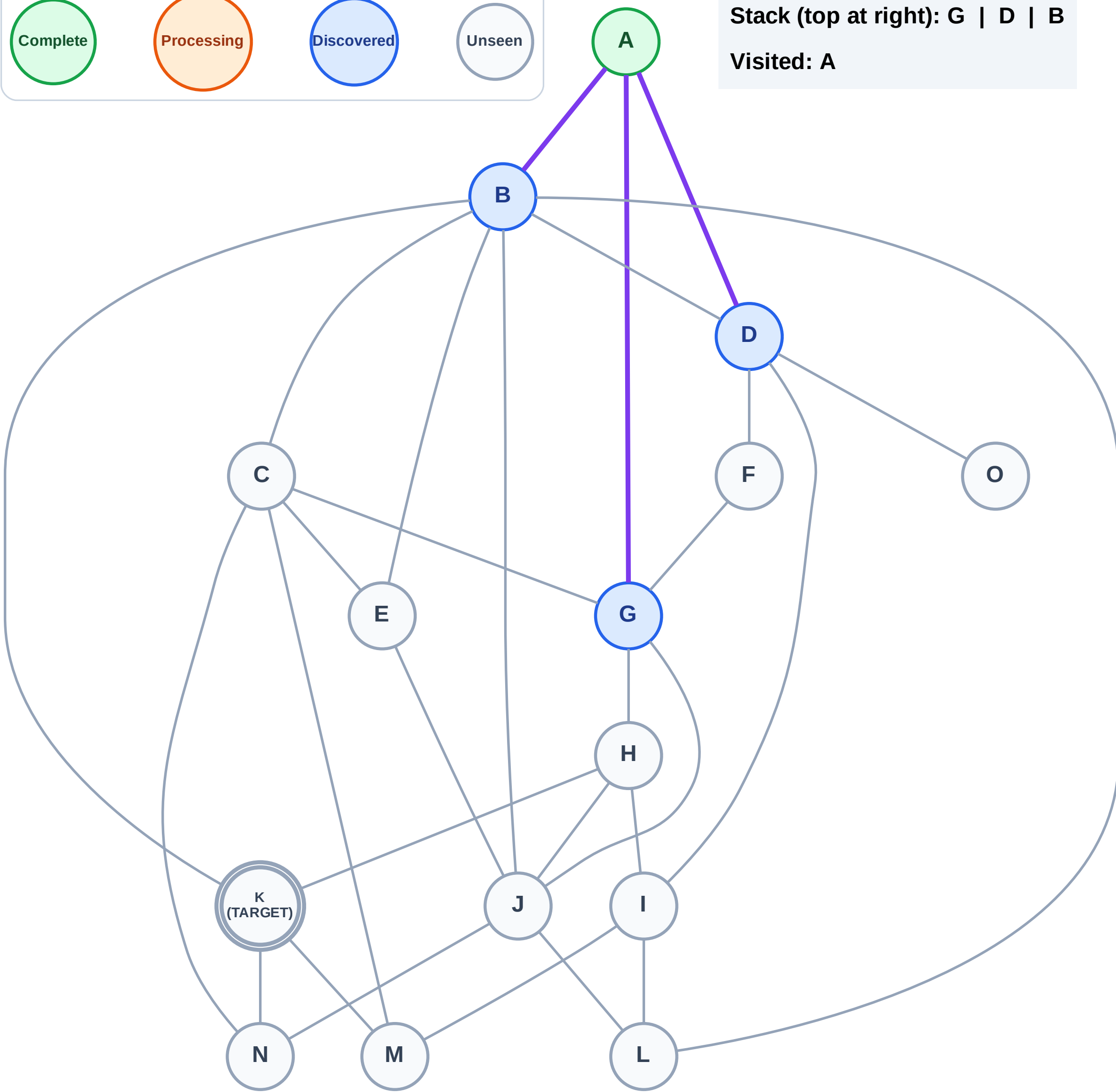
Discovered



Unseen

Stack (top at right): G | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

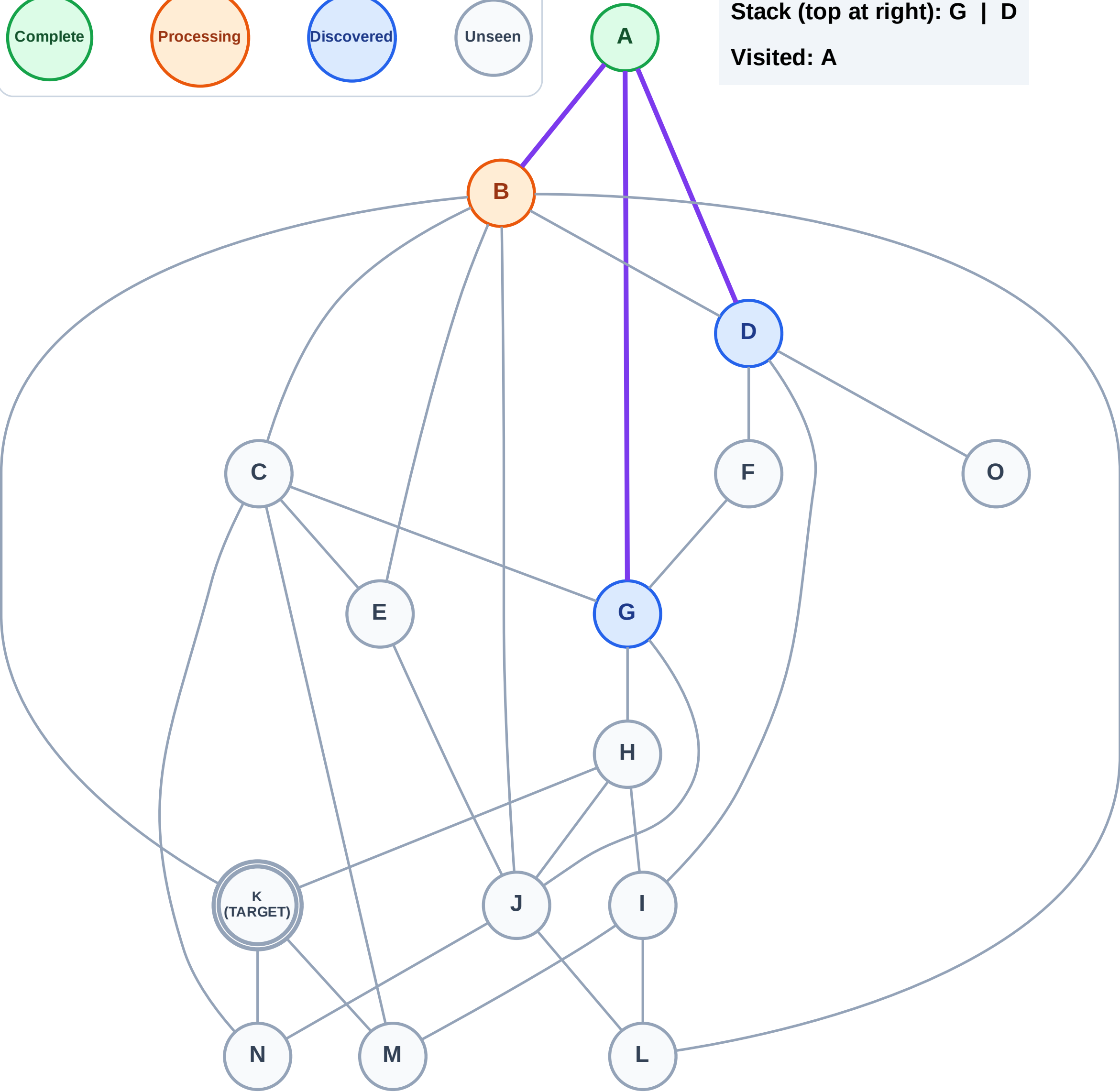
Complete

Processing

Discovered

Unseen

Stack (top at right): G | D
Visited: A



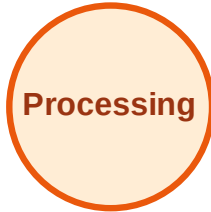
DFS Traversal

Step 5: Discover L, K, J, E, C from B

Legend



Complete



Processing



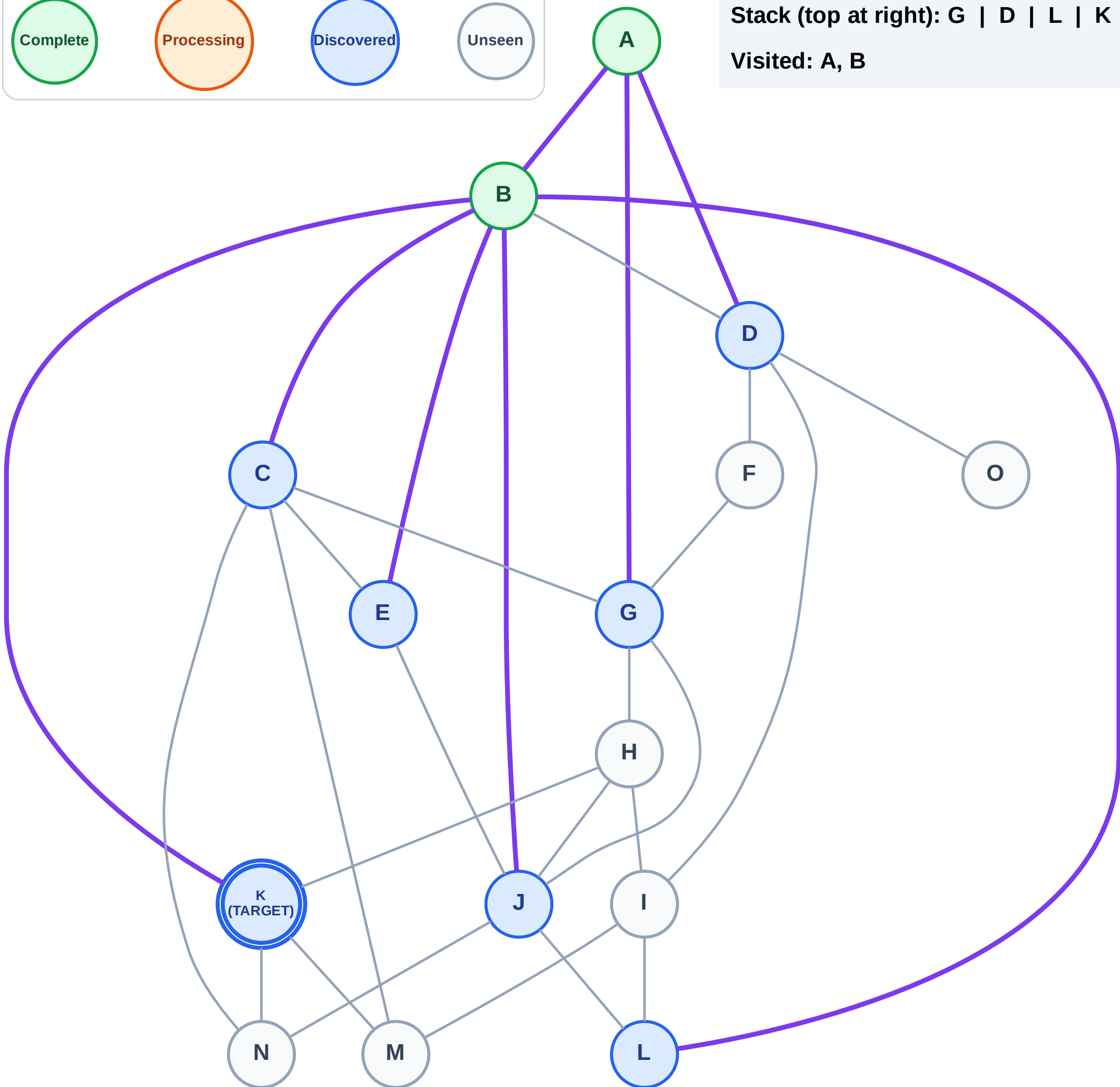
Discovered



Unseen

Stack (top at right): G | D | L | K | J | E | C

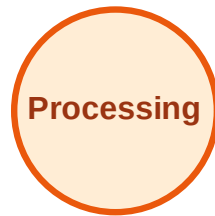
Visited: A, B



DFS Traversal

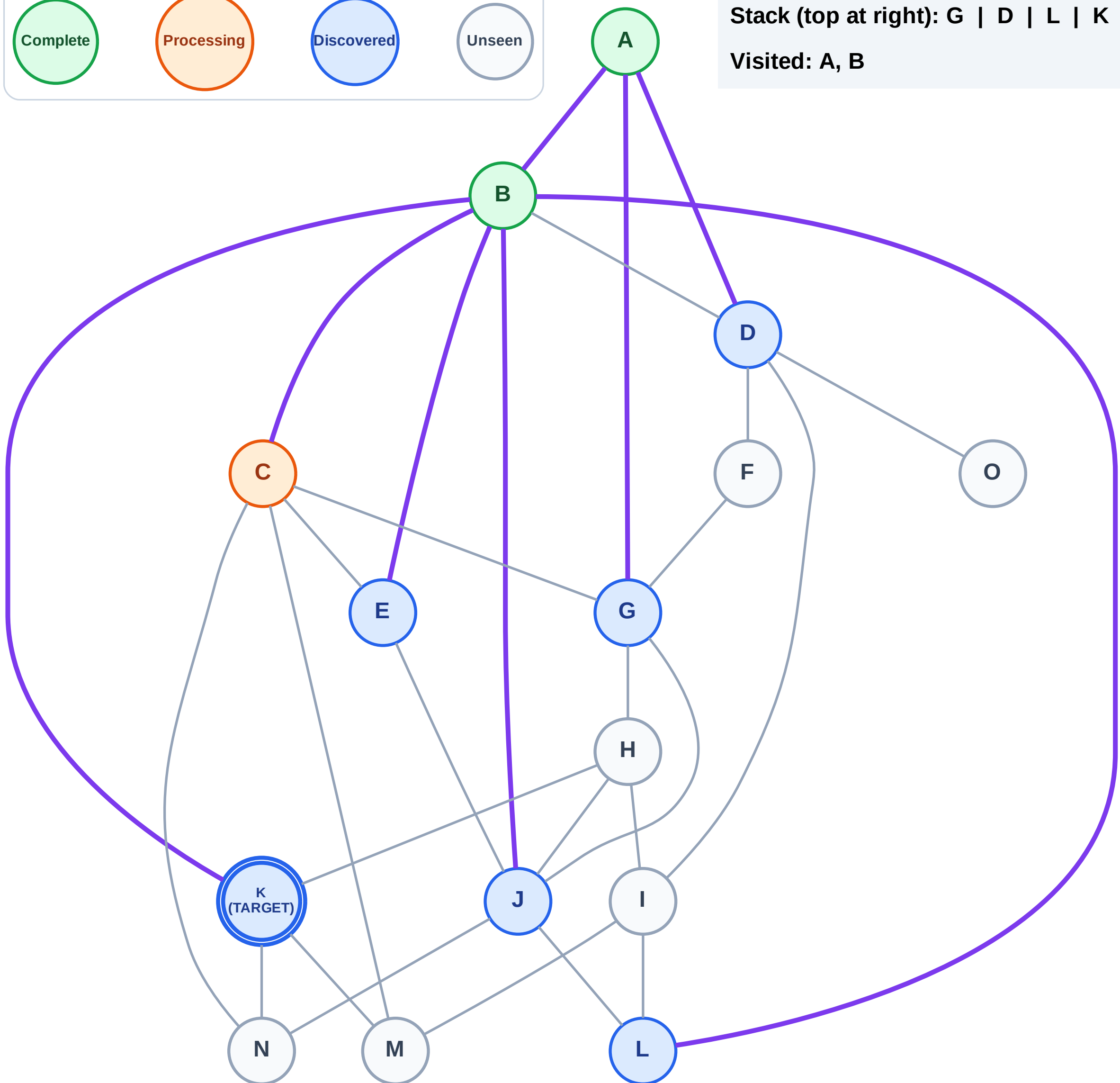
Step 6: Process node C

Legend



Stack (top at right): G | D | L | K | J | E

Visited: A, B



DFS Traversal

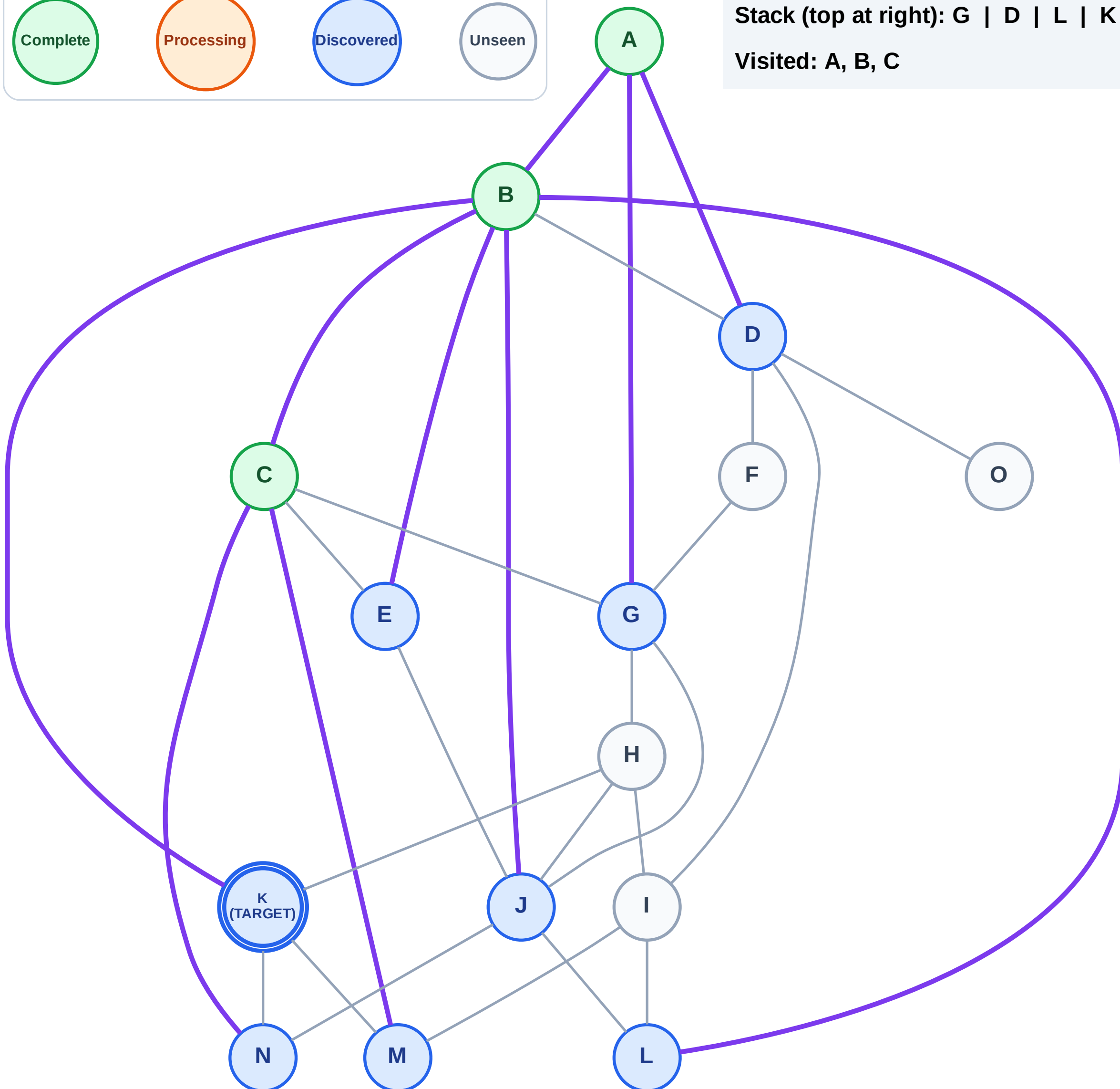
Step 7: Discover N, M from C

Legend



Stack (top at right): G | D | L | K | J | E | N | M

Visited: A, B, C



DFS Traversal

Step 8: Process node M

Legend

Complete

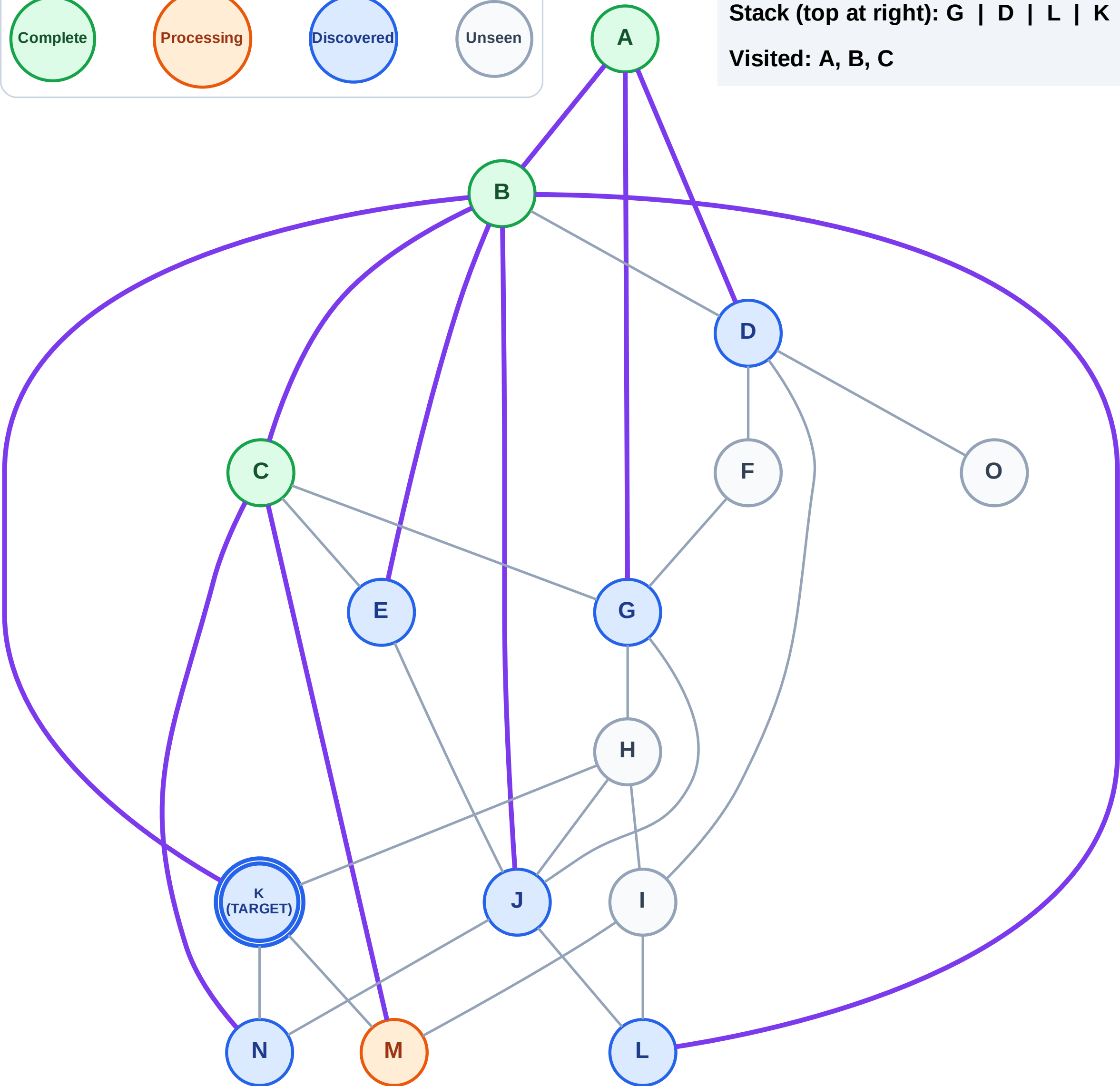
Processing

Discovered

Unseen

Stack (top at right): G | D | L | K | J | E | N

Visited: A, B, C



DFS Traversal

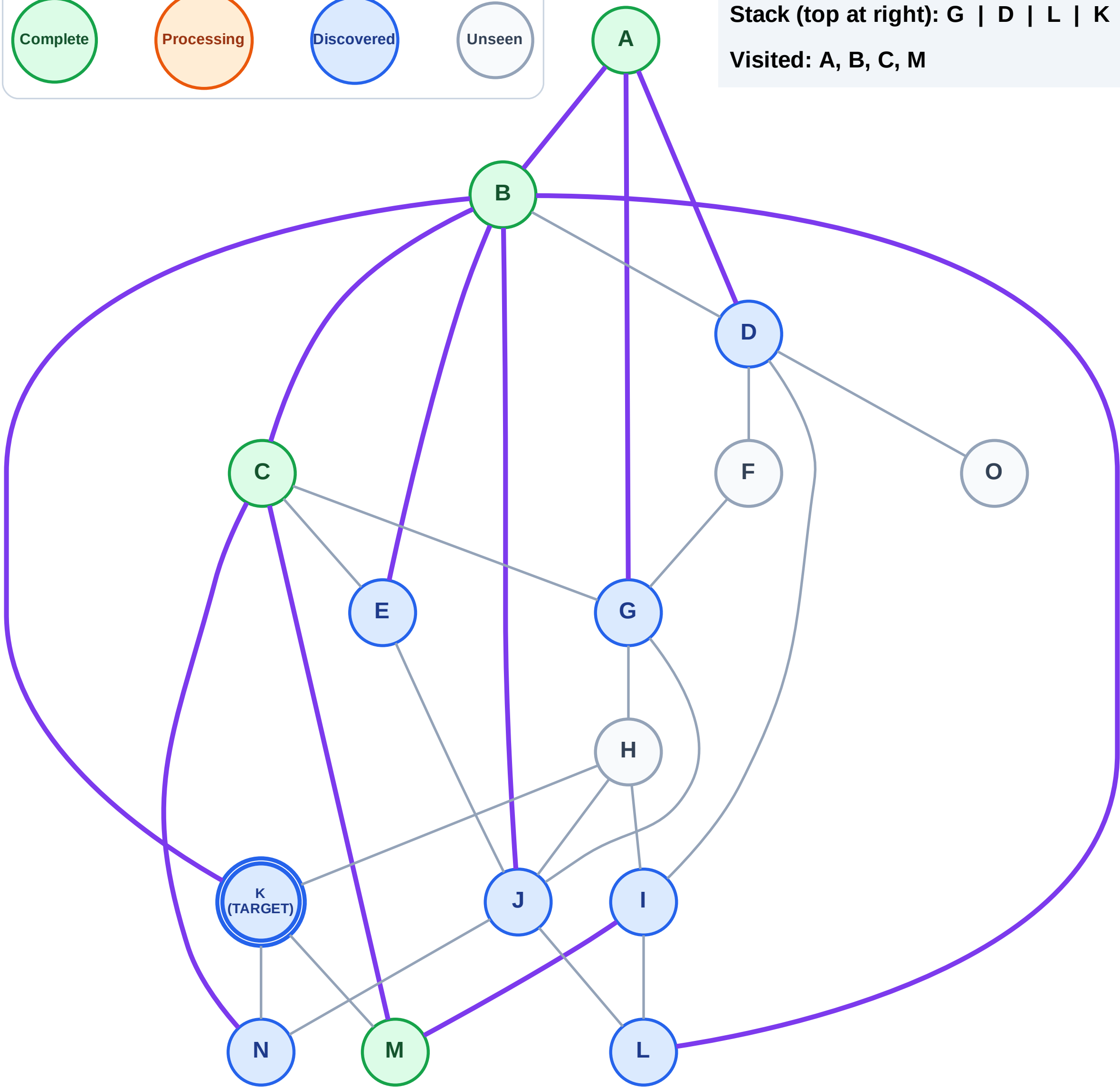
Step 9: Discover I from M

Legend



Stack (top at right): G | D | L | K | J | E | N | I

Visited: A, B, C, M



DFS Traversal

Step 10: Process node I

Legend

Complete

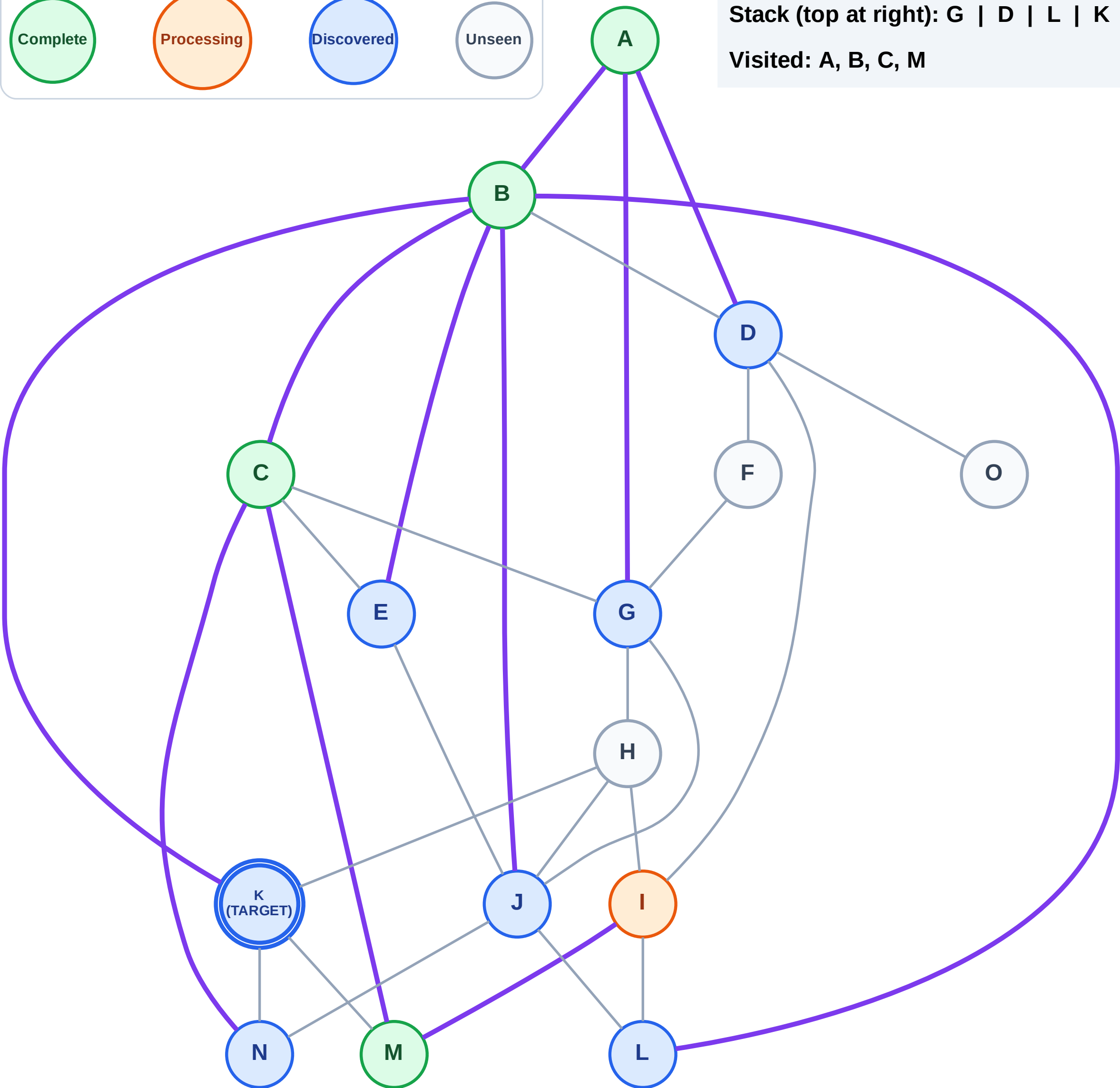
Processing

Discovered

Unseen

Stack (top at right): G | D | L | K | J | E | N

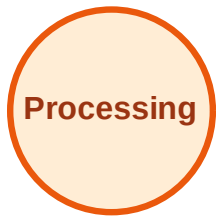
Visited: A, B, C, M



DFS Traversal

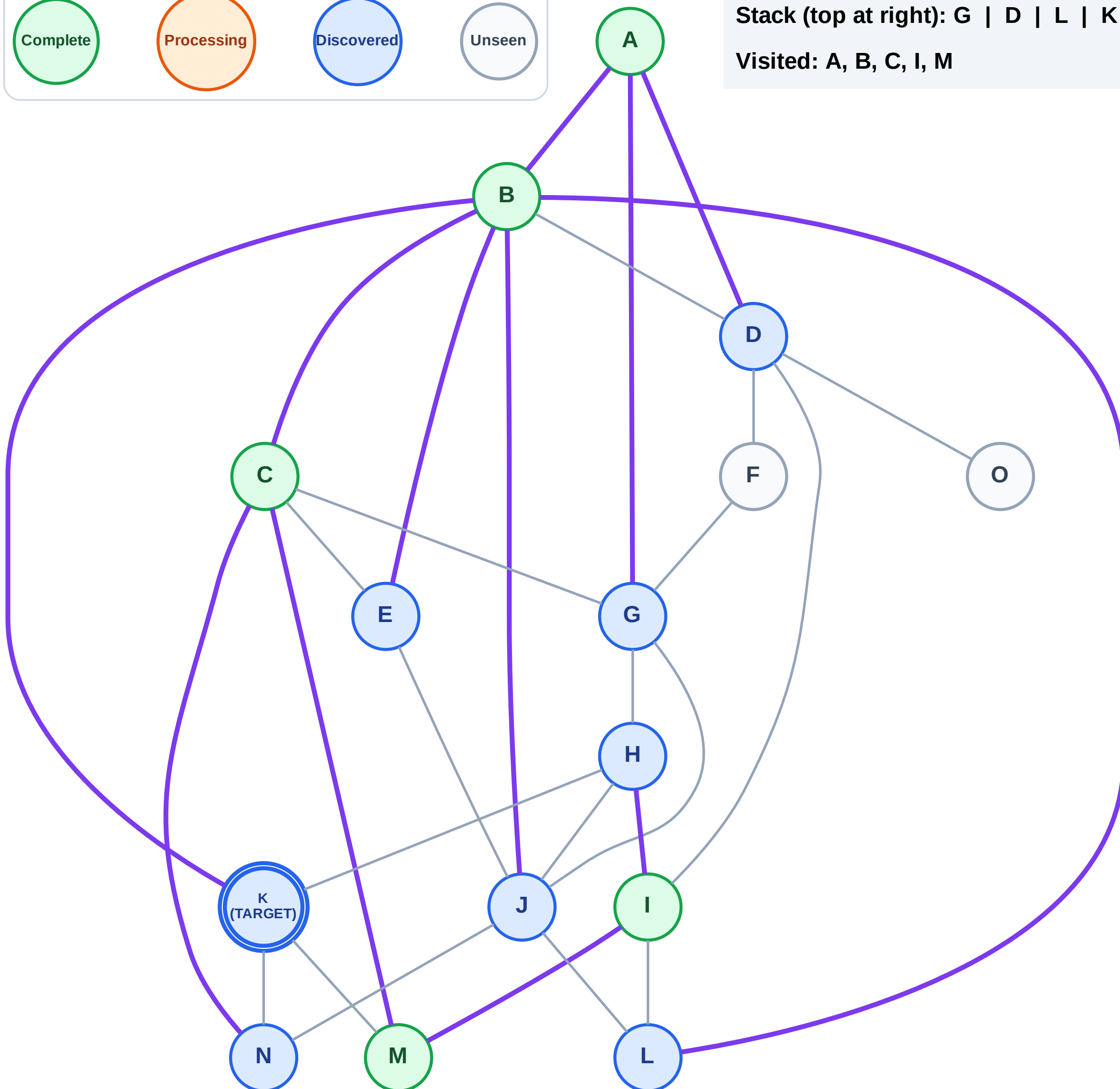
Step 11: Discover H from I

Legend



Stack (top at right): G | D | L | K | J | E | N | H

Visited: A, B, C, I, M



DFS Traversal

Step 12: Process node H

Legend

Complete

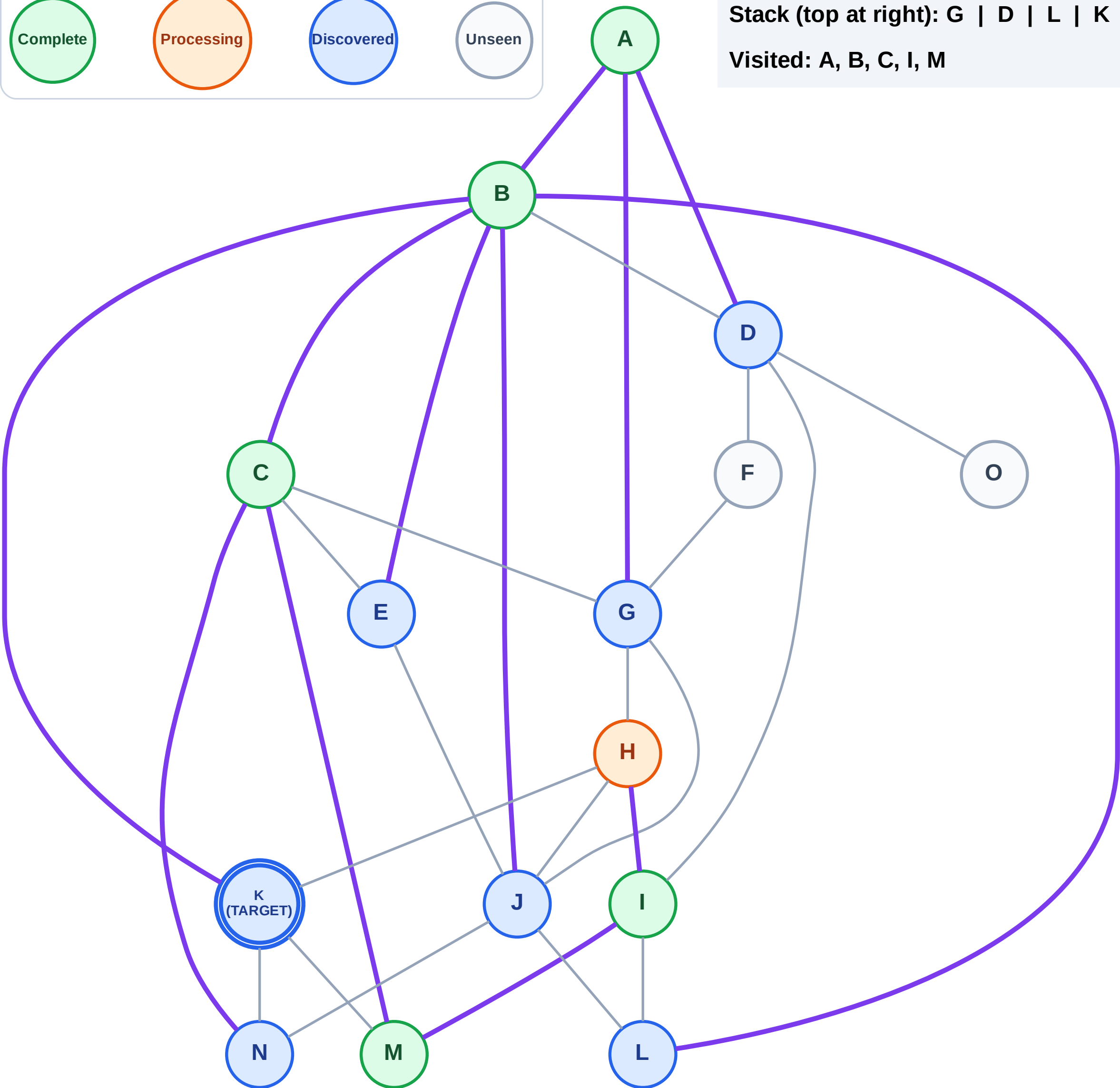
Processing

Discovered

Unseen

Stack (top at right): G | D | L | K | J | E | N

Visited: A, B, C, I, M



DFS Traversal

Step 13: No new neighbors from H

Legend

Complete

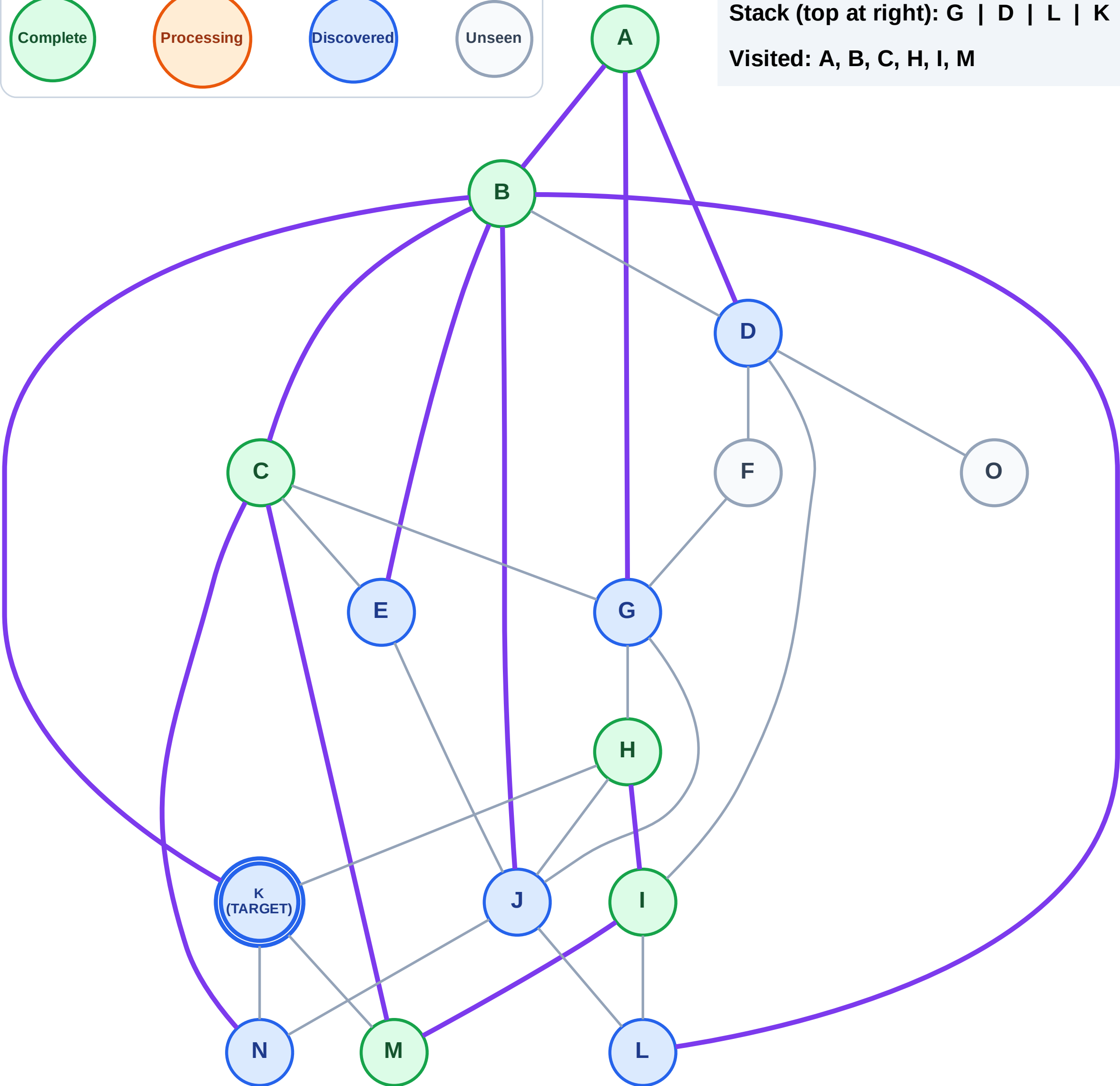
Processing

Discovered

Unseen

Stack (top at right): G | D | L | K | J | E | N

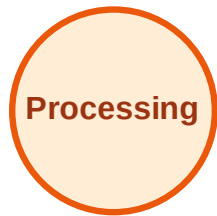
Visited: A, B, C, H, I, M



DFS Traversal

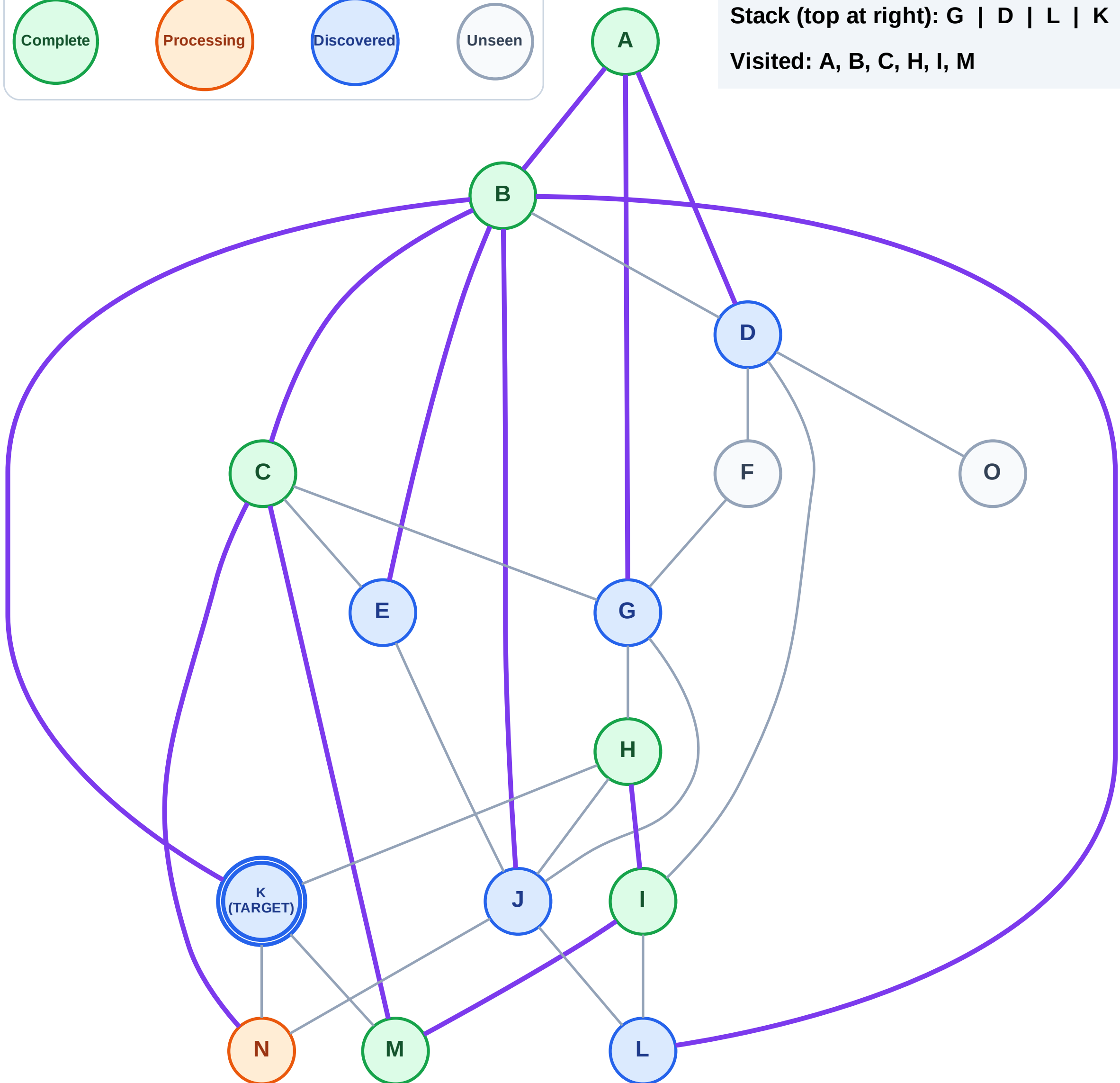
Step 14: Process node N

Legend



Stack (top at right): G | D | L | K | J | E

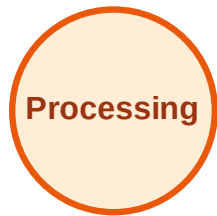
Visited: A, B, C, H, I, M



DFS Traversal

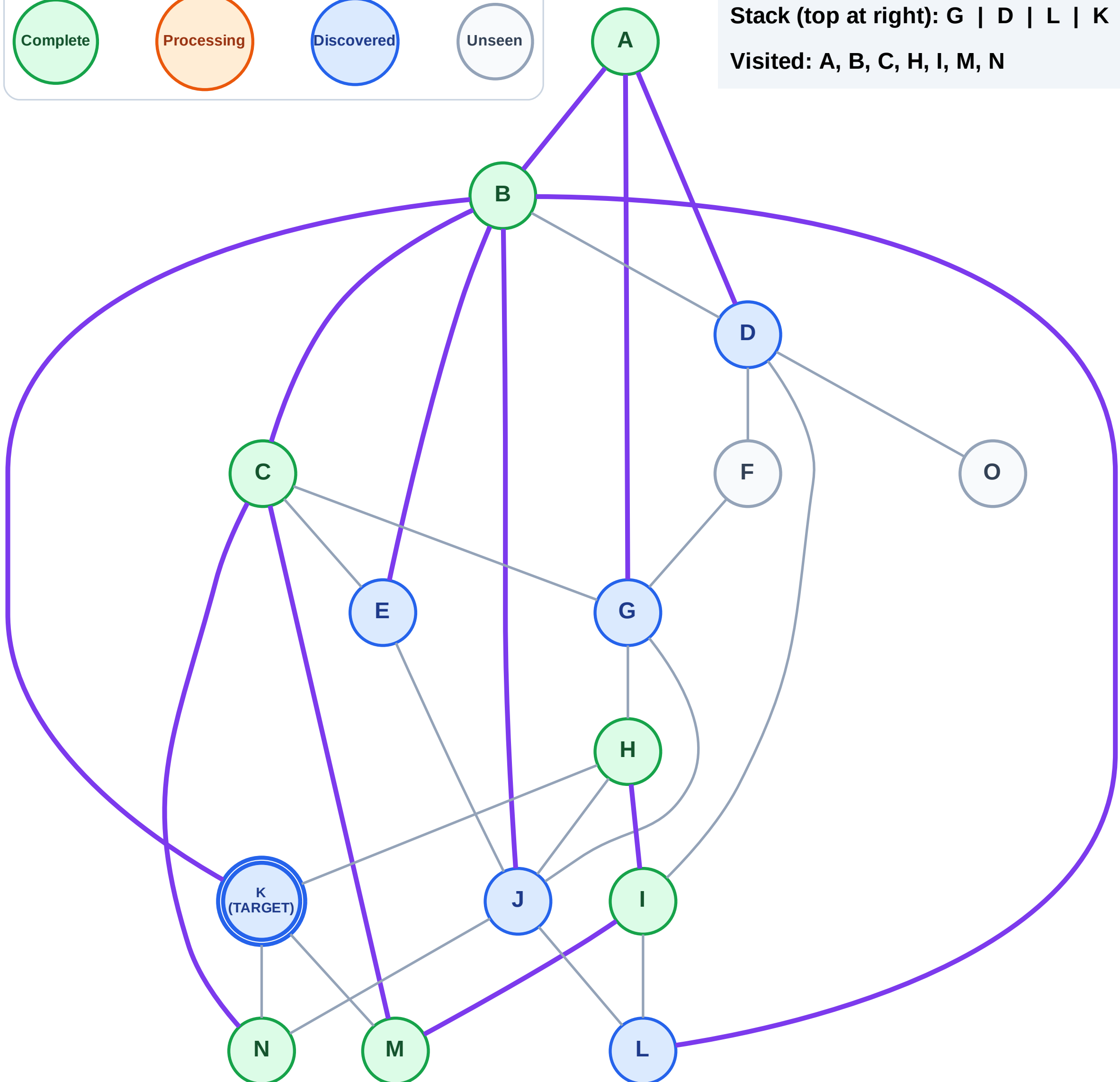
Step 15: No new neighbors from N

Legend



Stack (top at right): G | D | L | K | J | E

Visited: A, B, C, H, I, M, N



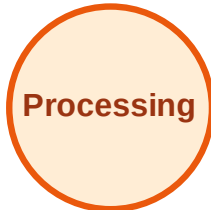
DFS Traversal

Step 16: Process node E

Legend



Complete



Processing



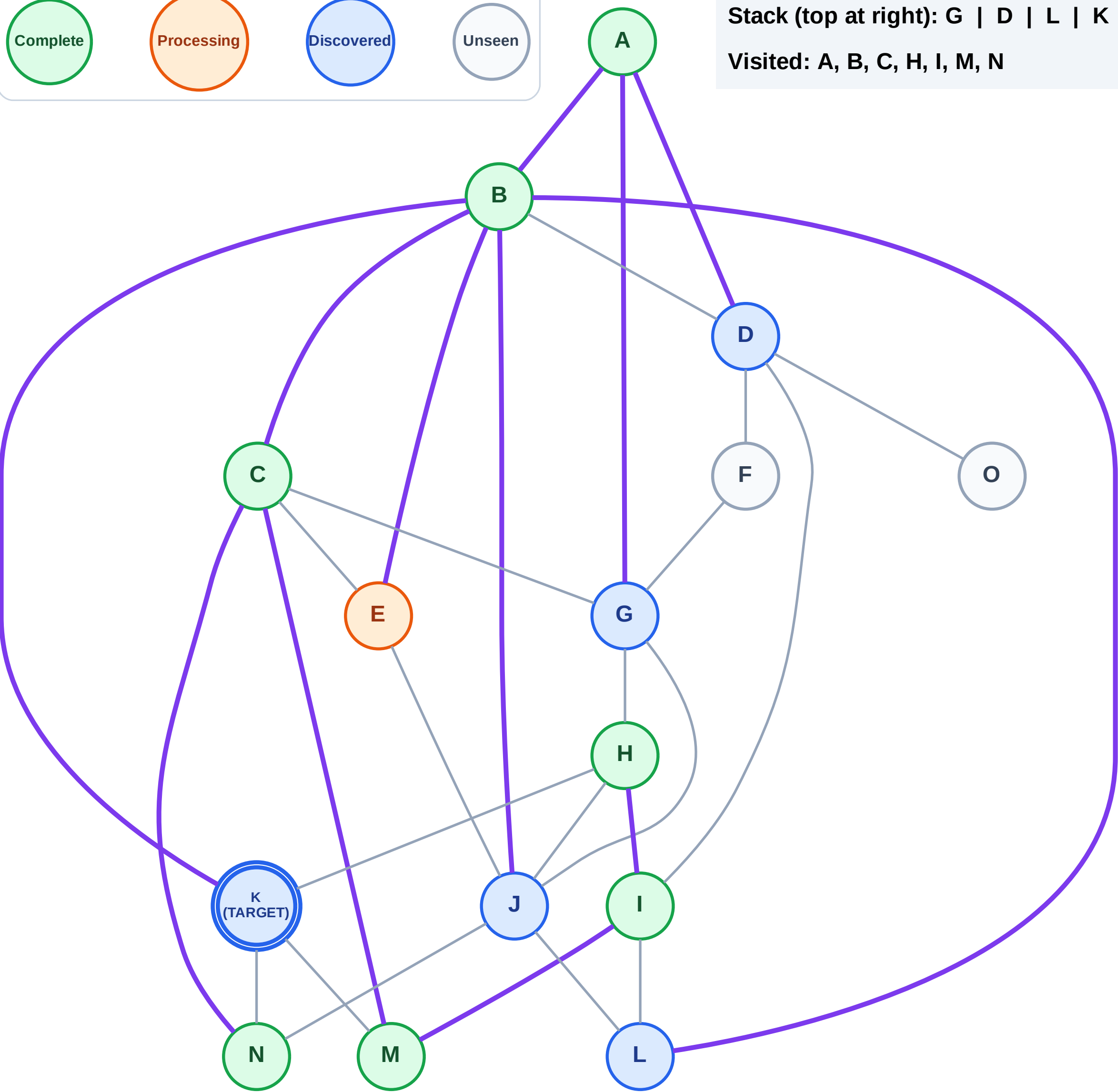
Discovered



Unseen

Stack (top at right): G | D | L | K | J

Visited: A, B, C, H, I, M, N



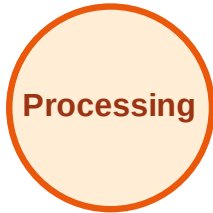
DFS Traversal

Step 17: No new neighbors from E

Legend



Complete



Processing



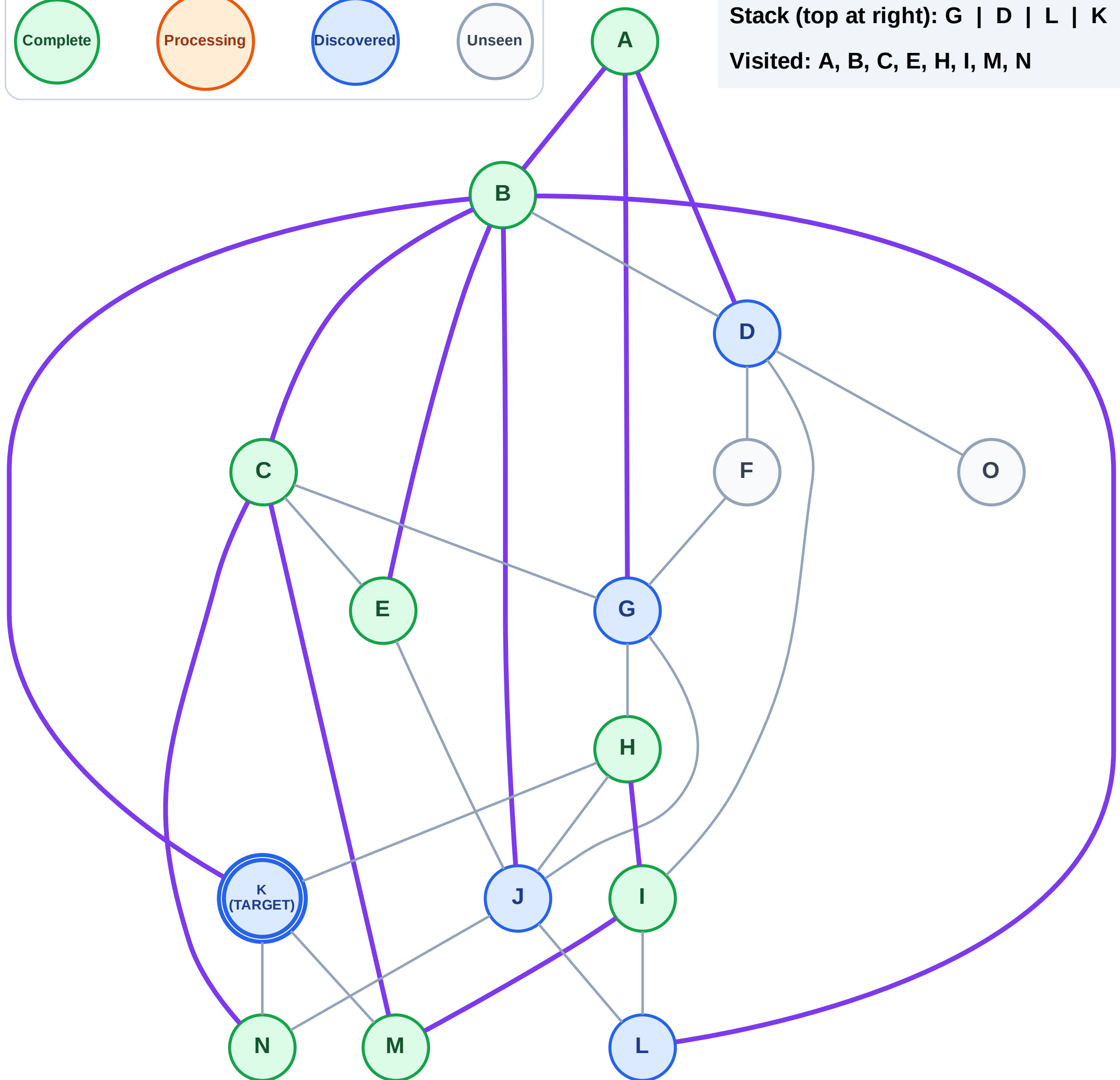
Discovered



Unseen

Stack (top at right): G | D | L | K | J

Visited: A, B, C, E, H, I, M, N



DFS Traversal

Step 18: Process node J

Legend

Complete

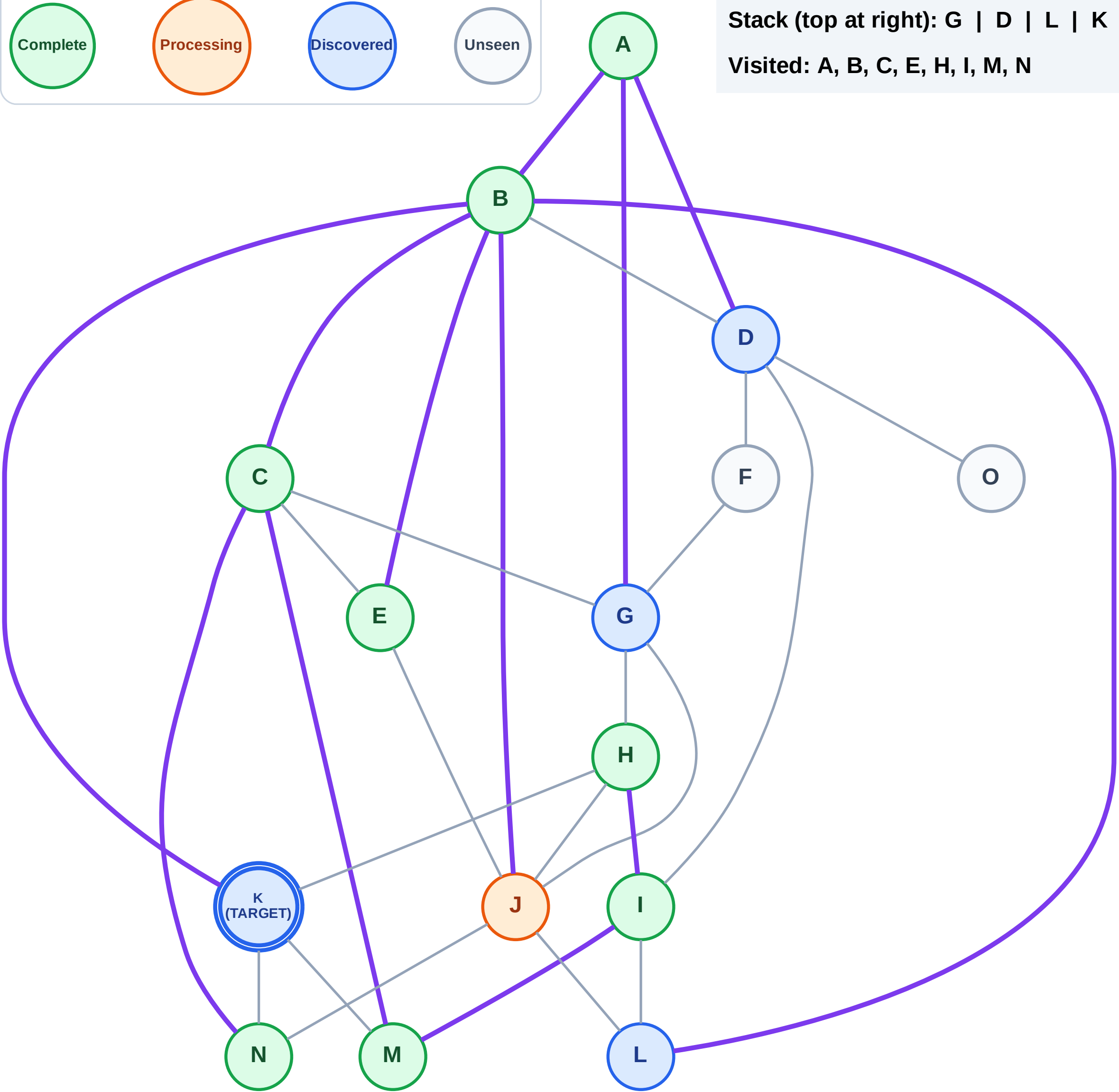
Processing

Discovered

Unseen

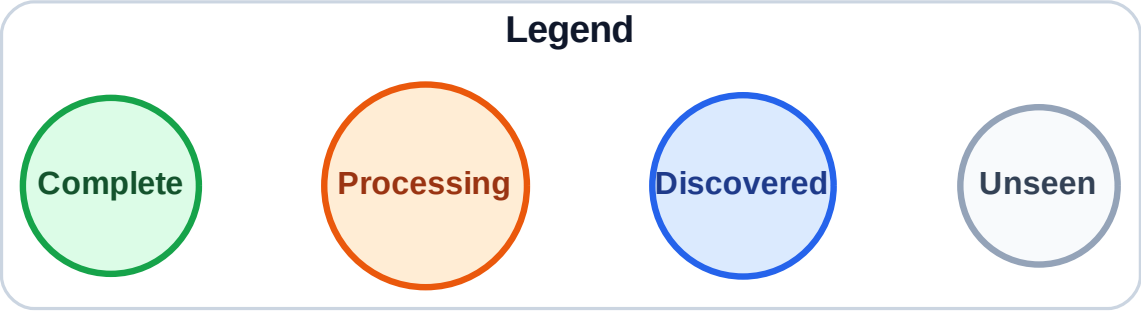
Stack (top at right): G | D | L | K

Visited: A, B, C, E, H, I, M, N

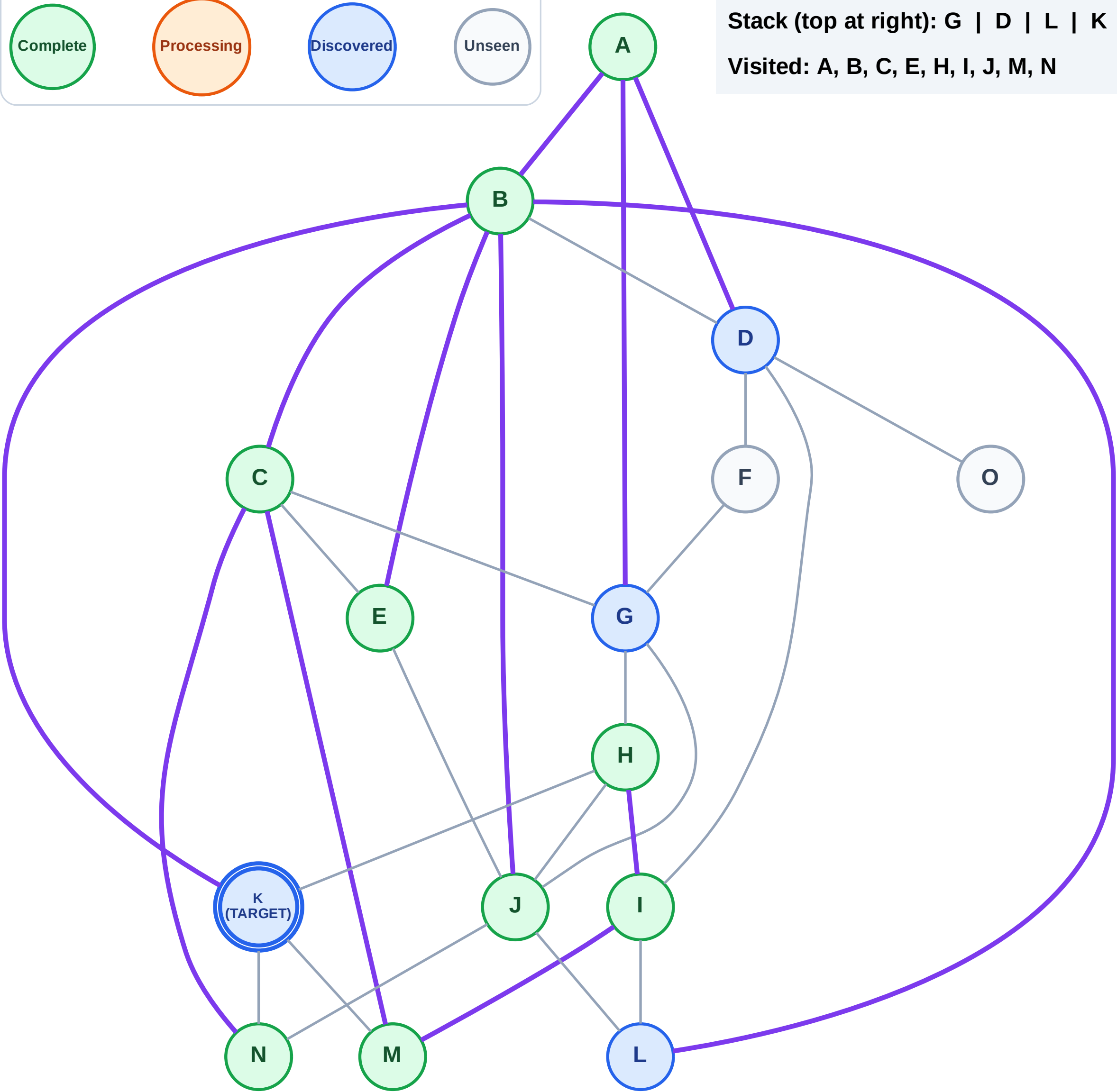


DFS Traversal

Step 19: No new neighbors from J



Stack (top at right): G | D | L | K
Visited: A, B, C, E, H, I, J, M, N

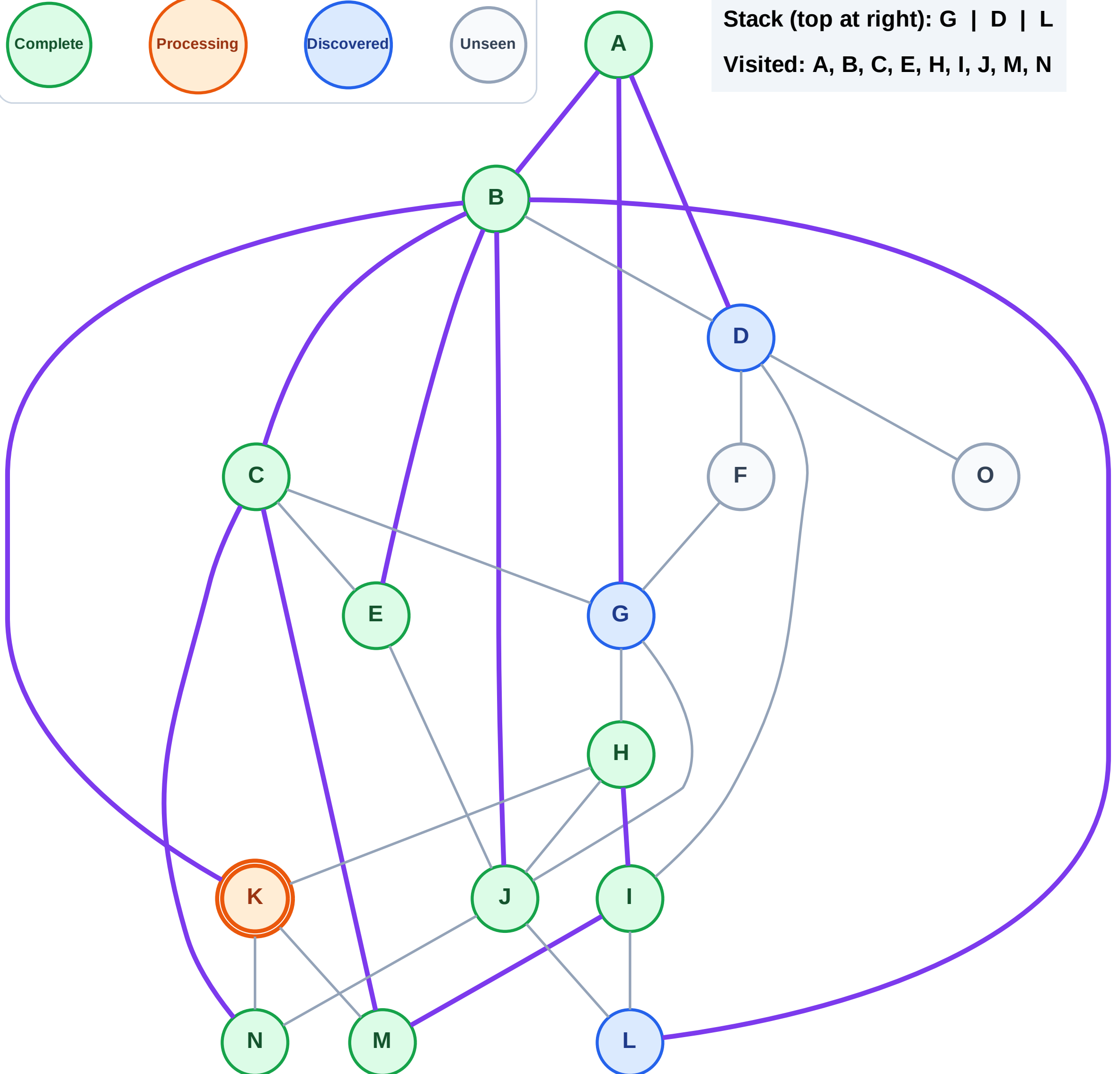


Step 20: Process node K

Legend

Unseen

Visited: A, B, C, E, H, I, J, M, N



DFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): G | D | L

Visited: A, B, C, E, H, I, J, K, M, N

